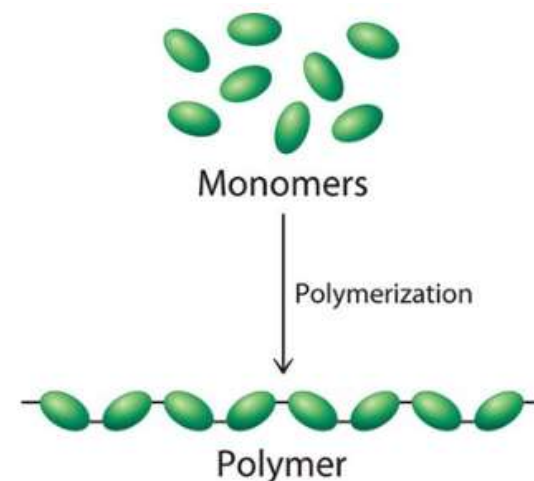
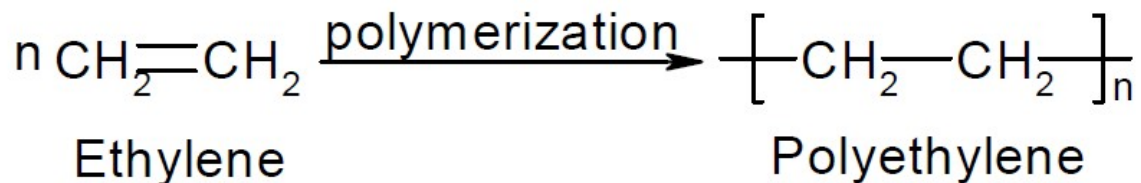


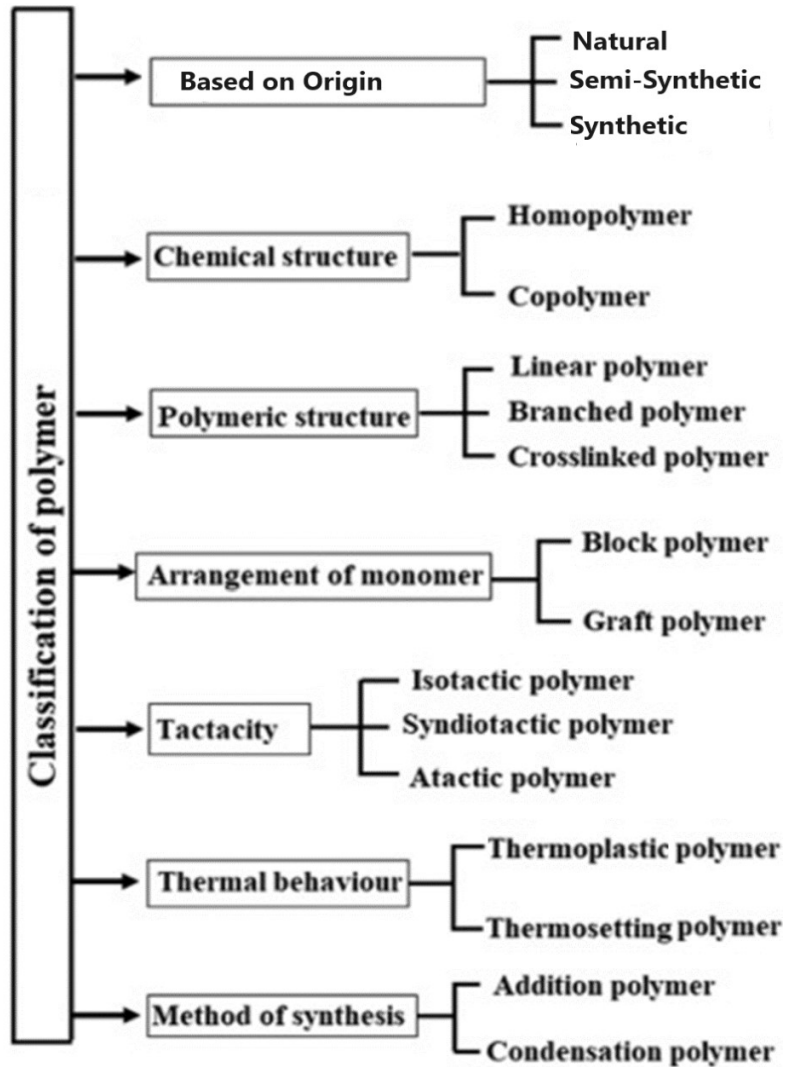
Unit 2 Polymer and Composites

INTRODUCTION

- Polymers are **macromolecules** formed by linking of many small molecules.
- The polymers are **giant molecules** with high molecular masses.
- The single repeating unit is called as **monomer**, and the resultant high molecular weight compound is called as polymer.
- For example, the monomer ethylene gets linked with many other ethylene molecules to form polyethylene.



CLASSIFICATION OF POLYMER



Classification based on source/origin

On the basis of sources of polymers they are classified in three types.

A. Natural polymers

- Polymers either obtained from plants or animal are called natural polymers. They are called plant and animal polymers.
- Ex. Cellulose, Jute, Lihen, Silk, Wool, Leather, RNA, DNA, Natural rubber.

B. Semi-synthetic polymers

- The polymers obtained by simple chemical treatment of natural fibers to improve their physical properties like lastrus nature, tensite strength are called semi-synthetic fibers.
- Ex: Acetate rayon, cuprammonium silk, viscous rayon.

C. Synthetic fibers

- The fibers obtained by polymerization of simple chemical molecules in laboratory are synthetic fibers.
- Ex. Nylon, terylene, polyethene, polystyrene, synthetic rubber, nylon, PVC, backlite, teflon, orion etc.

Classification Based on Chemical Structure



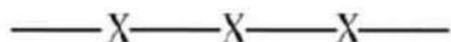
Homopolymers	Copolymers
The polymer formed by only one type of monomers is called homopolymer.	The polymer formed by more than one type of monomers is called copolymer.
Homopolymers are formed through addition polymerization.	Copolymers are formed through condensation polymerization.
Homopolymer consists of single species of repeating units.	Copolymer consists of two or more type of repeating units.
Homopolymer have single type of monomer species.	Copolymer have two or more types of monomer species.
Homopolymer often have a simple structure.	Copolymer have a complex structure.
Examples of homopolymer include PVC, Polyethylene, Polypropylene, Polystyrene, etc.	Examples of copolymer include Nylon 6,6, Dacron, SBR, etc.

Classification based on the structure of polymers

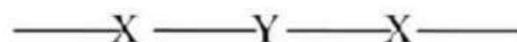
On the basis of structure they are classified in three types.

A. Linear polymers

- If the monomer units are joined in a linear fashion, polymer is said to be a linear polymer
- In these polymers monomers are linked with each other and form a long straight chain.
- These chains have no any side chains. Ex. Polyethene, PVC, Nylons, polyesters etc.
- Their molecules are closely packed and have high density, tensile strength, and melting point.



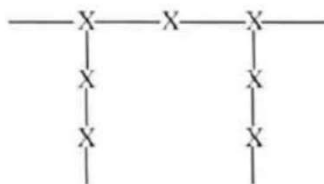
Linear Homopolymer



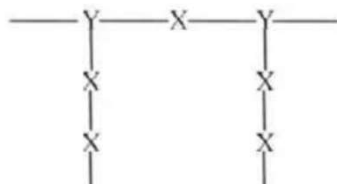
Linear Copolymer

B. Branched polymers

- When the monomer units are joined in branched manner, it is called branched polymer.
- They have a straight long chain with different side chains.
- Their molecules are irregularly packed hence they have low density, tensile strength and melting point. Ex. polypropylene (side chain $-\text{CH}_3$), amylopectin and glycogen.



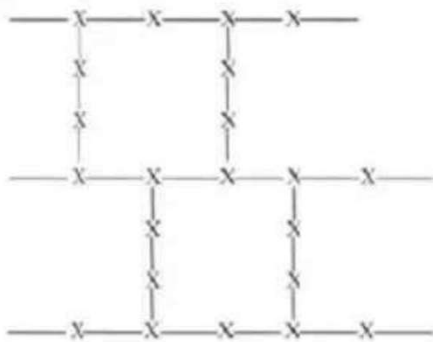
Branched Homopolymer



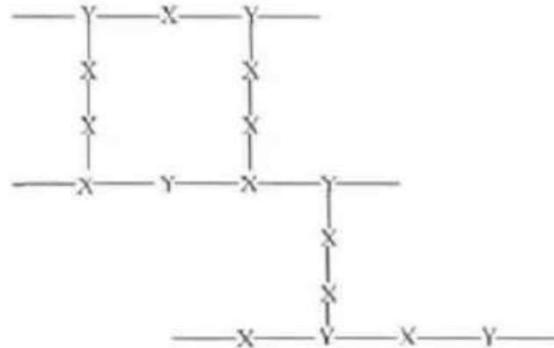
Branched Copolymer

C. Network or cross linked polymers

- A polymer is said to be a cross linked polymer, if the monomer units are joined together in a chained fashion
- In these monomeric units are linked together to constitute a three dimensional network.
- The links involved are called cross links. They are hard, rigid and brittle due to their network structure. Ex. Bakelite, maiamine, formaldehyde resins, vulcanized rubber etc.



Cross linked Homopolymer



Cross linked Copolymer

Classification Based on Arrangement of Monomers

Alternating



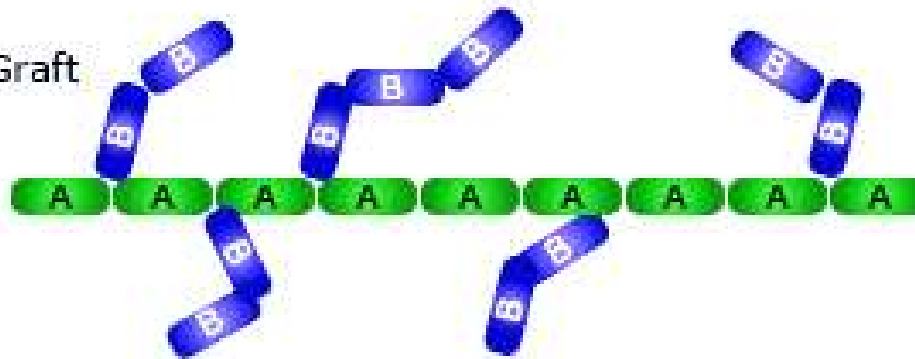
Block



Random



Graft



Classification based on thermal response

On the basis of thermal response, polymers can be classified into two groups

A. Thermoplastic polymers

- They can be softened or plasticized repeatedly on application of thermal energy, without much change in properties if treated with certain precautions.
- These are the polymers having intermolecular forces between elastomers and fibers.
- They are easily molded in desired shapes by heating and subsequent cooling at room temperature.
- They may be linear or branched chain polymers.
- They are soft in hot and hard on cooling.
- Ex. Polyolefins, nylons, sealing wax, linear polyesters and polyethers, PVC.

B. Thermosetting polymers

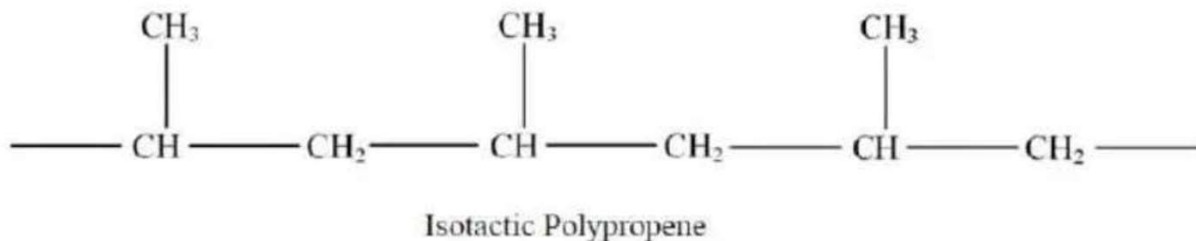
- Some polymers undergo certain chemical changes on heating and convert themselves into an infusible mass. The curing or setting process involves chemical reaction leading to further growth and cross linking of the polymer chain molecules and producing giant molecules.
- This polymer is hard and infusible on heating.
- These are not soft on heating under pressure and they are not remolded.
- These are cross linked polymers and are not reused.
- Ex. Bakelite, phenolic resins, urea, epoxy resins, diene rubbers, etc.

Classification based on tacticity

It may be defined as the geometric arrangement (orientation) of the characteristic group of monomer unit with respect to the main chain (backbone) of the polymers. On the basis of structure, polymer may be classified into three groups.

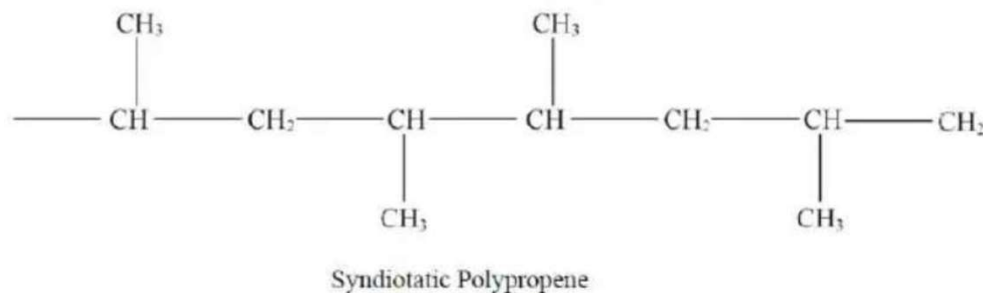
A. Isotactic polymer

- It is the type of polymer in which the characteristic group is arranged on the same side of the main chain.



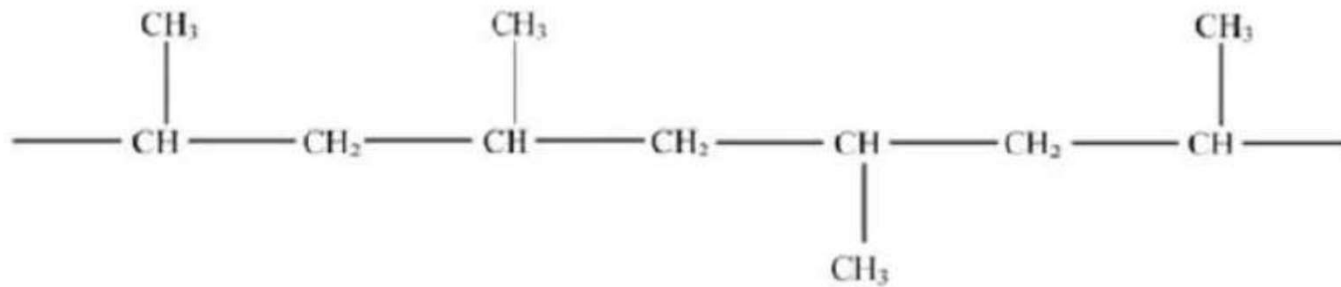
B. Syndiotactic polymer

- A polymer is said to be syndiotactic if the side group (characteristic group) are arranged in an alternate fashion.



C. Atactic polymer

- A polymer is said to be atactic, if the characteristic groups (side group) are arranged in irregular fashion (randomness) around the main chain.
- It has proper strength and more elasticity



Atactic Polypropene

Classification based on polymerization process

On this basis they are classified in two types.

A. Addition polymers

- The polymers formed by the addition of monomers repeatedly without removal of by products are called addition polymers.
- These polymers contain all the atoms of monomers hence they are integral multiple of monomer unit.
- The monomeric units are generally alkenes and its derivatives.
- Ex. Orion, teflon, polyethene, polypropylene, PVC.

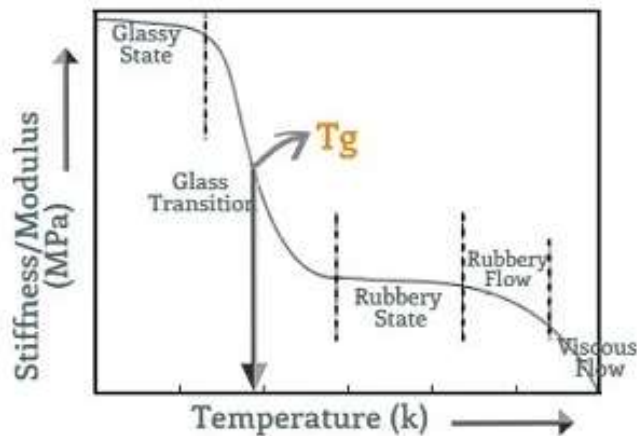
B. Condensation polymers

- They are formed by the combination of two monomers by removal of small molecules like water, alcohol or NH_3 .
- They have ester and amide linkage in their molecules.
- Their molecular mass is not the integral multiple of monomer units.
- Ex. Polyamides (Nylons), polyesters, polyurethans.

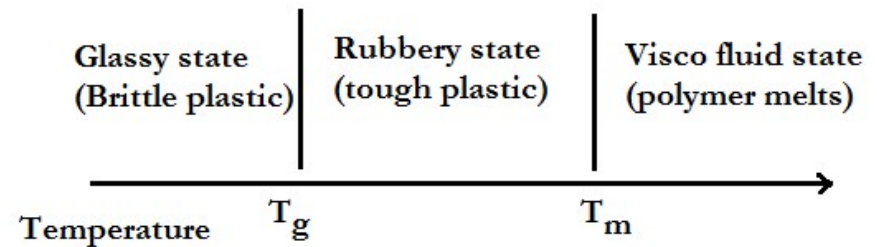
Glass Transition Temperature (T_g)

What is glass transition temperature (T_g)?

- T_g is the temperature at which an amorphous polymer changes from a hard/glassy state to a soft/leathery state, or vice versa.
- An important feature of **polymer behavior**.
- Changes in the **physical and mechanical properties**.



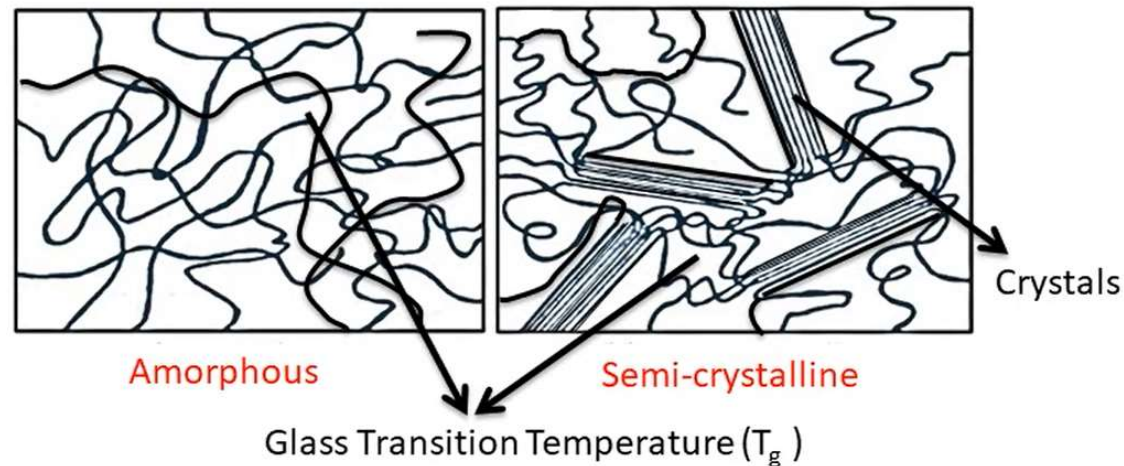
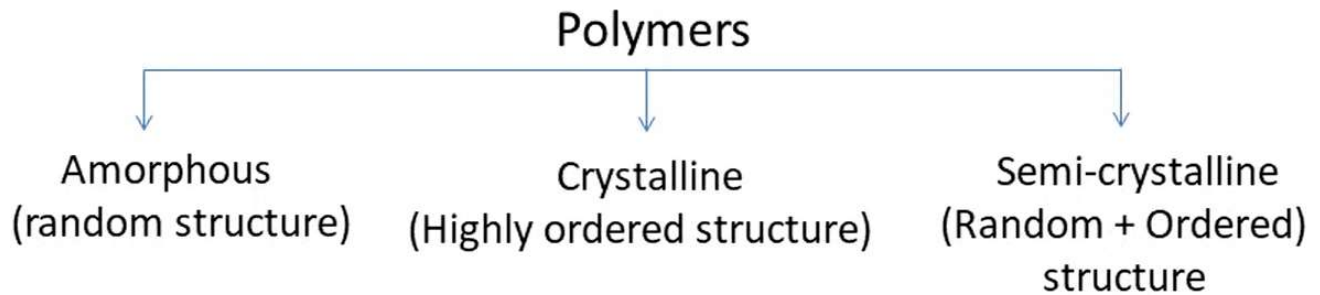
Glass transition temperature curve showing polymer transition



Below T_g : Due to lack of mobility, the polymers are hard and brittle like glass.

Above T_g : Due to some mobility, the polymers are soft and flexible like rubber.

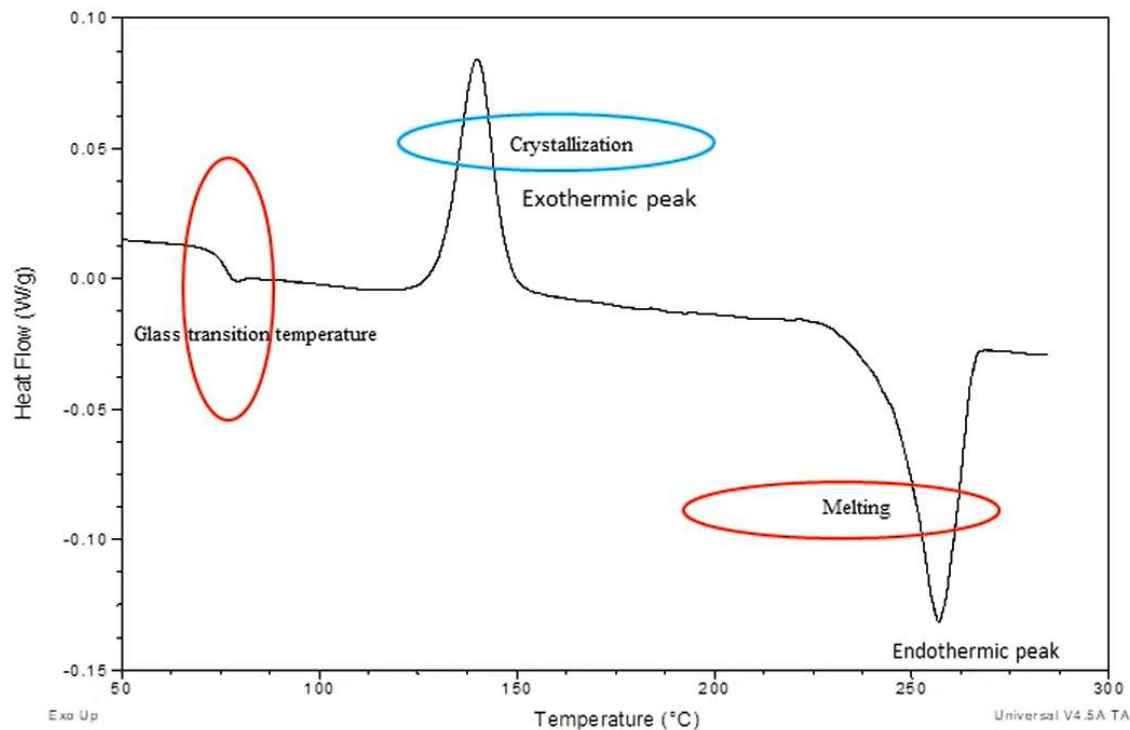
Glass Transition Temperature (T_g)



- T_g is a property of amorphous portion of a semi-crystalline solid.
- Higher the free volume lower the T_g .
- Higher the crystallinity higher the T_g .

Determination of Glass Transition Temperature (T_g)

Differential scanning calorimetry (DSC)



Heat Capacity: The amount of heat it takes to raise the temperature of the material by one degree.

Applications and Importance of Tg:

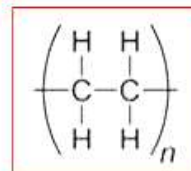
1. **Materials Behavior:** Tg impacts the behavior of polymers. Below Tg, the material is **glassy and rigid**, while above Tg, it becomes **rubbery or viscous**.
2. **Processing and Manufacturing:** Tg influences the **processing conditions** for polymers. For instance, during molding or extrusion, knowing Tg helps optimize temperature and cooling rates.
3. **Stability and Controlled Properties:** Many industries require materials with **stable and controlled Tg values**. These include aerospace, automotive, electronics, and packaging.
4. **Design Optimization:** Understanding Tg aids in designing materials with desired properties. For example, selecting a polymer with an appropriate Tg **ensures optimal performance in specific applications**.
5. **Thermal Expansion and Viscosity:** Tg affects **thermal expansion, viscosity**, and other physical characteristics of materials.

Factors Affecting Glass Transition Temperature (T_g):

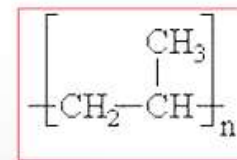
1) **Chain Flexibility and Rigidity:** Stiffening groups in the polymer chain reduce the flexibility of the chain and raise the value of T_g. In other words, greater the intrinsic chain flexibility, smaller will be the T_g.

Example: T_g (PP) T_g (PE)

Since substitution of H in PE by CH₃ group in PP impedes the rotation about a C-C single bond. Thus, T_g is affected by the nature of substitution



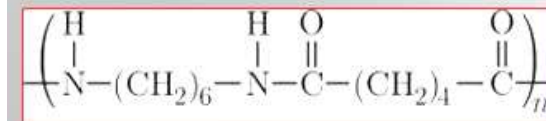
Polyethylene (PE)



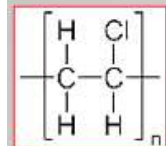
Polypropylene (PP)

2) Intermolecular Forces: T_g (PE) < T_g (PVC or T_g (Nylon 66)

Stronger intermolecular forces due to dipole forces, H-bonding etc. decreases the mobility of the chain lead to a higher T_g. PVC has stronger intermolecular forces than polypropylene because of the dipole-dipole forces from the C-Cl bond.



Nylon 66



Polyvinyl chloride (PVC)

3) Molecular Weight: Glass transition temperature, **T_g is directly proportional to molecular weight of the polymer.**

4) Pendant Groups: The influence of pendant groups on the glass transition temperature is somewhat more complicated. Bulky pendant groups, such as a benzene ring, can catch on neighboring chains like a "fish hook" and **restrict rotational freedom and can increase T_g.**

5) Cross-Linking: The presence of cross-links between chains restricts rotational motion and hence, **greater the degree of cross-linking, the higher the T_g.**

6) Plasticizers: Plasticizers are low molecular weight compounds added to plastics to increase their flexibility and workability. They weaken the intermolecular forces between the polymer chains **as it increases the mobility and decrease T_g.**

7) Copolymerization: **Random copolymers have lower T_g's** since it tends to promote the disorder, reduce molecular packing and reduces the inter chain forces of attraction.

Glass transition temperature

→ directly proportional to molecular weight of polymer.

→ directly proportional to degree of cross links

→ directly proportional to intermolecular force.

→ directly proportional to number of side groups

→ inversely proportional to flexibility.

→ inversely proportional to plasticization.

Degree of Polymerization (Dp)

Degree of Polymerization (Dp)



No. of monomer units present
in a polymer

$$DP = \frac{M}{m}$$

Where **Dp** is the degree of polymerization,
M is the molar mass of the polymer and
m is the molar mass of the monomer unit.

- High degrees of polymerization indicate long polymer chains and complex structures.
- Low degrees suggest short chains and simpler structures

Polymer Molecular Weight

$$\text{Number Average Molecular Weight} = \overline{M}_n = \frac{\sum N_i M_i}{\sum N_i}$$

$$\text{Weight Average Molecular Weight} = \overline{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i}$$

$$\text{Z Average or Size Average Molecular Weight} = \overline{M}_z = \frac{\sum N_i M_i^3}{\sum N_i M_i^2}$$

$$\text{Viscosity Average Molecular Weight} = M_v = \left[\frac{\sum N_i M_i^{a+1}}{\sum N_i M_i} \right]^{1/a}$$

$$PD = \frac{\overline{M}_w}{\overline{M}_n}$$

What is Number Average Molecular Weight?

- Number average molecular weight is the **mole fraction of molecules** in a polymer sample.
- It is a way of determining the **molecular mass** of a polymer.
- It gives the **average of the molecular masses** of the individual macromolecules.
- In other words, it is the **total weight of the sample divided by the number of molecules in the sample**.

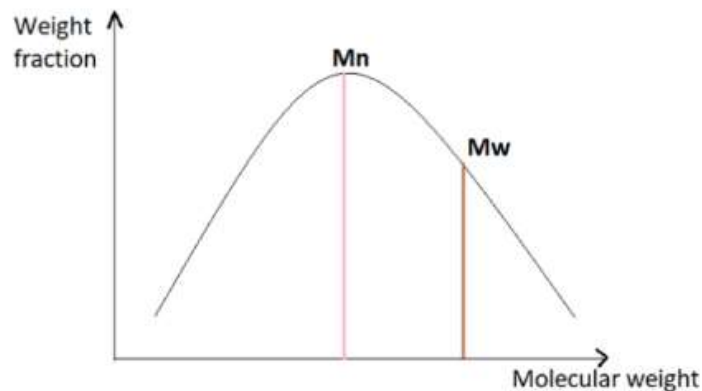


Figure: $M_w > M_n$

$$\text{Number Average Molecular Weight} = \overline{M}_n = \frac{\sum N_i M_i}{\sum N_i}$$

What is Weight Average Molecular Weight?

- Weight average molecular weight is the **weight fraction of molecules** in a polymer sample.
- It is another way of determining the **molecular mass of a polymer**.
- It gives the average of the **molecular masses of the individual macromolecules** in the polymer sample.

$$\text{Weight Average Molecular Weight} = \overline{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i}$$

Where, n=Moles of molecules (n₁+n₂+n₃+-----n_i)

M=Molecular weight of each molecule

Difference Between Number Average and Weight Average Molecular Weight

- The key difference between **number average** and **weight average** molecular weight is that the number average molecular weight refers to the **mole fraction of molecules in a polymer sample** whereas the weight average molecular weight is the **weight fraction of molecules** in a polymer sample.
- We can determine the **number average molecular weight** using **gel permeation chromatography, viscometry** and **colligative methods** such as vapour pressure osmometry
- We can determine the **weight average molecular weight** using **static light scattering, small angle neutron scattering, X-ray scattering, and sedimentation velocity**.

Numericals:

- ① A polymer has the following molar mass distribution.

No. of molecules	molar mass ($\text{g}\cdot\text{mol}^{-1}$)
50	5000
75	6000

calculate number average molar mass ($\bar{M}_n$) & weight average molar mass ($\bar{M}_w$) of the polymer.

→ Given $n_1 = 50$, $n_2 = 75$
 $M_1 = 5000$, $M_2 = 6000$

$$\begin{aligned}\text{number average molar mass } (\bar{M}_n) &= \frac{\sum n_i m_i}{\sum n_i} \\&= \frac{n_1 m_1 + n_2 m_2}{n_1 + n_2} \\&= \frac{50(5000) + 75(6000)}{50 + 75} \\&= \frac{250000 + 450000}{125} \\&= 5600 \text{ g}\cdot\text{mol}^{-1}\end{aligned}$$

$$\begin{aligned}
 \text{Weight average molar mass } (\bar{M}_w) &= \frac{\sum n_i m_i^2}{\sum n_i m_i} \\
 &= \frac{n_1 m_1^2 + n_2 m_2^2}{n_1 m_1 + n_2 m_2} \\
 &= \frac{50 (5000)^2 + 75 (6000)^2}{(50 \times 5000) + (75 \times 6000)} \\
 &= 5642 \text{ g.mol}^{-1}.
 \end{aligned}$$

$$PD = \frac{\bar{M}_w}{\bar{M}_n}$$

A polymer sample contains 100, 200, 300 & 400 molecules having molecular mass 1000, 2000, 3000 & 4000 respectively. Calculate the number average & wt. average molecular masses of the polymer.

$$\begin{aligned}\bar{M}_w &= \frac{\sum n_i m_i^2}{\sum n_i m_i} = \frac{n_1 m_1^2 + n_2 m_2^2 + n_3 m_3^2 + n_4 m_4^2}{100 \times 1000 + 200 \times 2000 + 300 \times 3000 + 400 \times 4000} \\ &= \frac{1.1 \times 10^6}{30,0000} = 3333.33 \text{ g.mol}^{-1}\end{aligned}$$

$$\begin{aligned}\bar{M}_n &= \frac{\sum n_i m_i}{\sum n_i} = \frac{n_1 m_1 + n_2 m_2 + n_3 m_3 + n_4 m_4}{100 + 200 + 300 + 400} \\ &= \underline{\underline{3000 \text{ g.mol}^{-1}}}\end{aligned}$$

$$PD = \frac{\bar{M}_w}{\bar{M}_n}$$

③ A polymer has the following molar mass distribution

No. of molecules	Molar mass (g/mol)
35	2000
65	4000
85	6000
95	8000

calculate $\bar{M}_n$ & $\bar{M}_w$.

MODE OF POLYMERIZATION

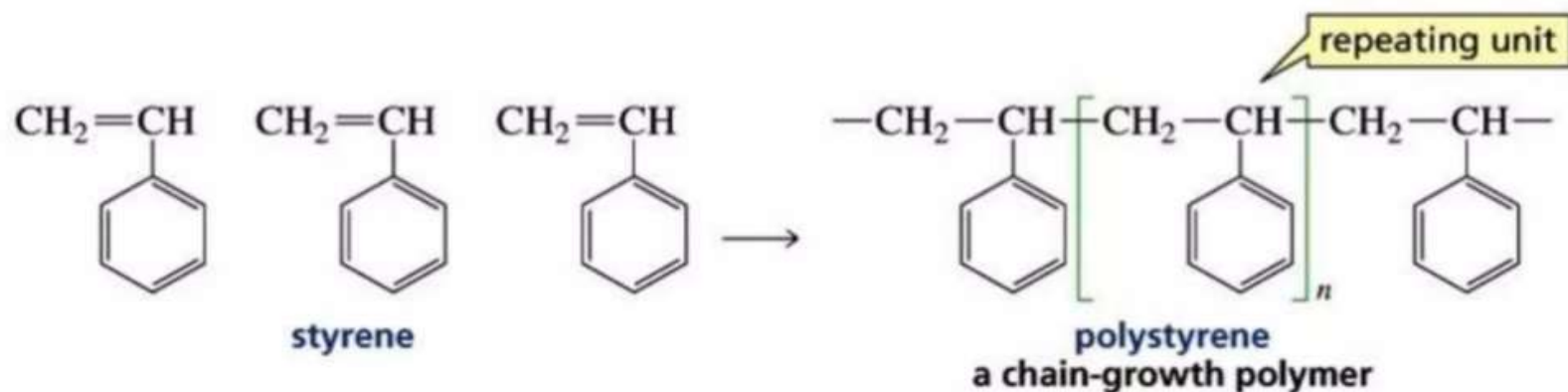
There are two modes of Polymerization:-

❖ Chain Growth Polymerization

❖ Step Growth Polymerization

CHAIN GROWTH POLYMERIZATION

- Chain growth polymerization is the process of **repeated addition** of monomers that possess double or triple bonds to form **polymers**.
- In which the **high molecular weight polymer** is formed.
- Also the **polymer yield** gradually increases with time.
- It is also known as **addition polymerization**.



EXAMPLE OF CHAIN GROWTH POLYMERIZATION



Table 28.1 Some Important Chain-Growth Polymers and Their Uses

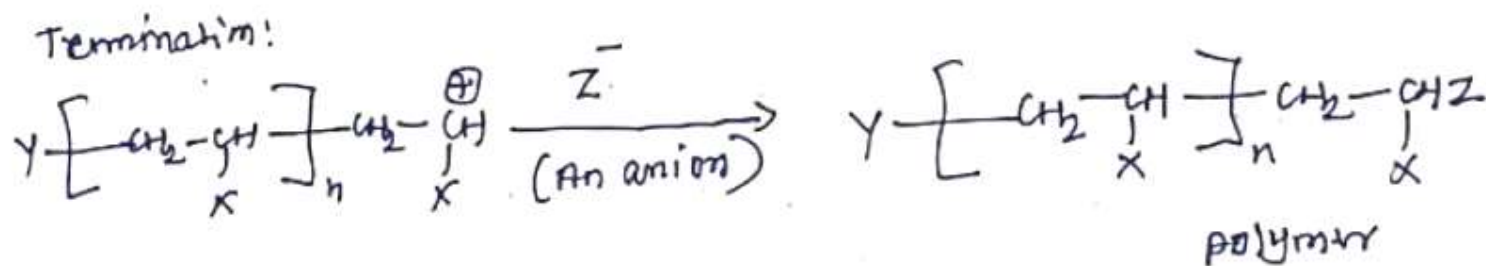
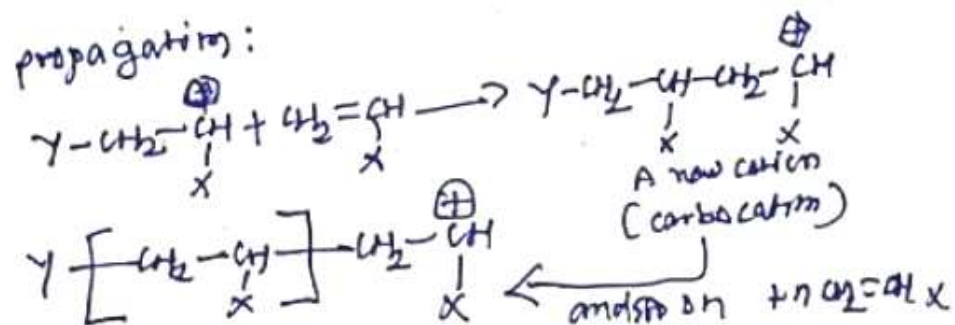
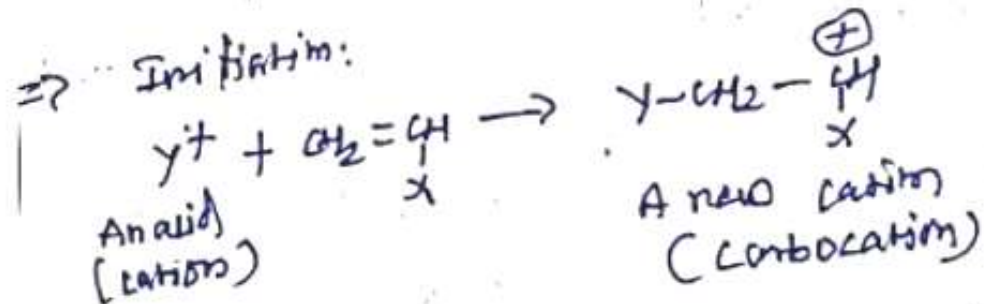
Monomer	Repeating unit	Polymer name	Uses
$\text{CH}_2=\text{CH}_2$	$-\text{CH}_2-\text{CH}_2-$	polyethylene	film, toys, bottles, plastic bags
$\text{CH}_2=\underset{\text{Cl}}{\text{CH}}$	$-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-$	poly(vinyl chloride)	"squeeze" bottles, pipe, siding, flooring
$\text{CH}_2=\text{CH}-\text{CH}_3$	$-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-$	polypropylene	molded caps, margarine tubs, indoor/outdoor carpeting, upholstery
$\text{CH}_2=\underset{\text{C}_6\text{H}_5}{\text{CH}}$	$-\text{CH}_2-\underset{\text{C}_6\text{H}_5}{\text{CH}}-$	polystyrene	packaging, toys, clear cups, egg cartons, hot drink cups
$\text{CF}_2=\text{CF}_2$	$-\text{CF}_2-\text{CF}_2-$	poly(tetrafluoroethylene) Teflon [®]	nonsticking surfaces, liners, cable insulation
$\text{CH}_2=\underset{\text{C}\equiv\text{N}}{\text{CH}}$	$-\text{CH}_2-\underset{\text{C}\equiv\text{N}}{\text{CH}}-$	poly(acrylonitrile) Orlon [®] , Acrilan [®]	rugs, blankets, yarn, apparel, simulated fur
$\text{CH}_2=\underset{\text{COCH}_3}{\text{C}}-\text{CH}_3$	$-\text{CH}_2-\underset{\text{COCH}_3}{\overset{\text{CH}_3}{\text{C}}}-$	poly(methyl methacrylate) Plexiglas [®] , Lucite [®]	lighting fixtures, signs, solar panels, skylights
$\text{CH}_2=\underset{\text{OCCH}_3}{\text{CH}}$	$-\text{CH}_2-\underset{\text{OCCH}_3}{\text{CH}}-$	poly(vinyl acetate)	latex paints, adhesives

Mechanism

It can be done by 4 Mechanisms:

- Cationic Polymerization
- Anionic Polymerization
- Free Radical Polymerization
- Coordination Polymerization or Ziegler-Natta Polymerization

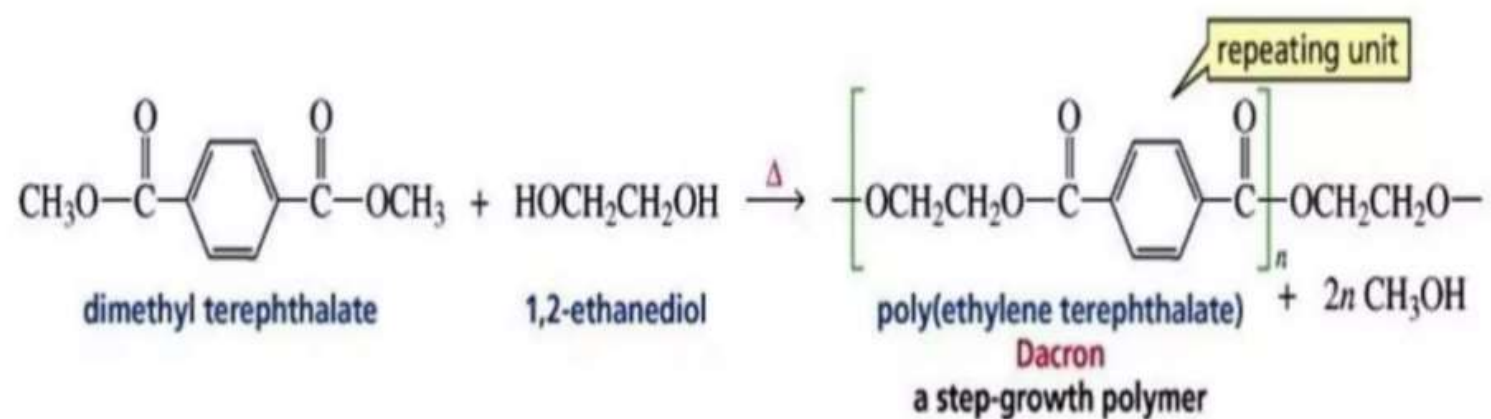
Cationic Polymerization Mechanism:



STEP GROWTH POLYMERIZATION

- Step-growth polymerization refers to a type of polymerization mechanism in which bi-functional or multifunctional monomers react to form first dimers, then trimers, longer oligomers and eventually long chain polymers.
- It is also known as condensation polymerization.
- Condensation polymerization is a process that involves repeated condensation reactions between two different bi-functional or tri-functional monomers.
- In condensation polymerization some product such as H_2O , NH_3 etc. are released.

STEP GROWTH POLYMERIZATION



The types of functional groups which are most imp. in cond. polymⁿ.

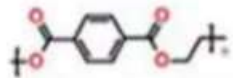
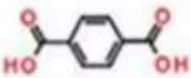
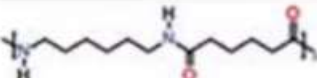
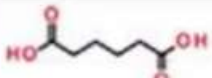
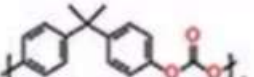
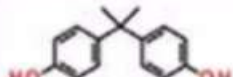
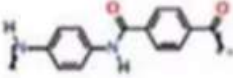
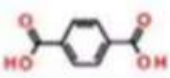
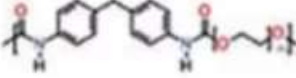
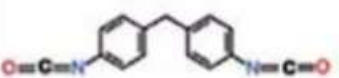
1 st group	2 nd group	product	Example
Hydroxyl -OH	Carboxyl -COOH	Polyester -OOC-	Polyethylene tere- phthalate (terylene)
Amino -NH ₂	—n—	Polyamide -NH-CO-	Nylon 6,6
Hydroxyl -OH	Isocyanate OCN-	Polyurethane -OC-NH-	Spandex fibre

If monomer contains 3 functional groups



Cross-linked structure polymer.

EXAMPLE OF STEP GROWTH POLYMERIZATION

Polymer	Monomers	Uses
 Polyethylene terephthalate (PET)	  Terephthalic acid Ethylene glycol	Bottles, films (MYLAR), aluminized films for food packaging and insulation, yarn for clothing (Terylene), etc.
 Nylon 6,6	  Adipic acid 1,6-hexamethylene diamine	Fibre for carpets, bags (Cordura), molded parts for automobile applications, ropes and nets, etc.
 Polycarbonate	  Bisphenol A Phosgene	Tough transparent sheets used for green houses, airplane/train windows, safety goggles, spectacle lenses, electronic applications, CD's etc.
 Kevlar	  Terephthalic acid 1,4-phenylenediamine	One of the strongest synthetic fibres; used in the manufacture of bullet-proof vests; also used to make fibre-reinforced composites.
 Polyurethane	  Methylene diisocyanate (MDI) Polyethylene glycol	Representative structure; variety of diols (polyols) and diisocyanates are used to tailor the properties. Used as rigid foams for packaging, thermal insulation and for flexible foams in upholstery; coatings and finishes; and thermoplastic elastomers.

Difference between Addition and Condensation Polymerization

S.No.	Addition polymerization	Condensation polymerization
1	It is also called chain growth polymerization.	It is also called step growth polymerization.
2	It takes place only in monomers containing multiple bonds.	It takes place in monomers containing functional groups.
3	The concentration of monomer decreases steadily throughout reaction.	The concentration of monomer disappears early in the reaction.
4	Elimination of simple molecule does not take place.	Elimination of simple molecule like H_2O , NH_3 , HCl etc. are essential.
5	High molecular weight polymer is formed at once	The molecular weight of polymer rises steadily throughout the reaction.
6.	Polyethylene, polystyrene, polypropene, poly (vinyl chloride) etc., are its examples.	Examples are terylene (dacron), nylon6, 6, nylon 6 etc.

Commercial Polymers

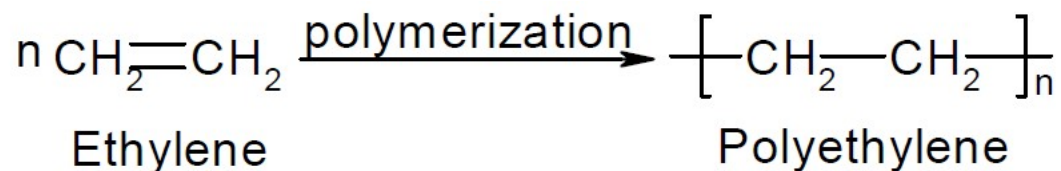
- 1. POLYETHYLENE (PE)**
- 2. POLYPROPYLENE (PP)**
- 3. POLYVINYL CHLORIDE (PVC)**
- 4. POLYSTYRENE (PS)**
- 5. PHENOL FORMALDEHYDE (PF)**
- 6. EPOXY RESIN**

Commercial Polymers

1. Polyethylene (PE):

Synthesis:

- PE is obtained by the polymerization of **ethylene**.
- The gas is first **liquefied under high pressure** (upto 1500 atm) and then pumped into a heated pressure **vessel**, maintained at **150 to 250°C**.
- By catalytic effects of traces of **oxygen** present, ethylene is polymerized into polyethylene.
- A **Waxy solid** is formed at the bottom of the vessel.



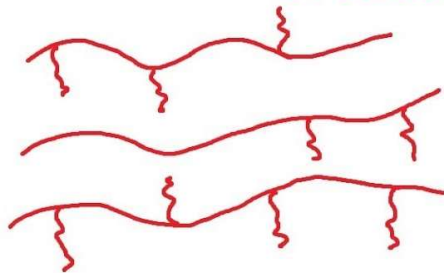
By using free radical initiator → LDPE is obtained

By using ionic catalyst → HDPE is obtained.



LDPE

Branched



Low density
0.91-0.94 g/cm³

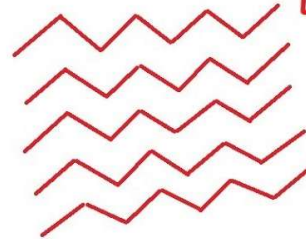
mp

~115°C

Low crystallinity (50-60%),

HDPE

Linear



High density
0.95-0.97 g/cm³

better packing density

~135°C

High crystallinity (>90%)

Properties:

1. Depending upon the density, they may be LDPE and HDPE.
2. If we use free radical initiator, LDPE is the product while use of ionic catalysts results in the formation of HDPE.
3. It is a rigid, waxy white solid. Translucent.
4. It is permeable to many organic solvents.
5. It crystallizes easily.

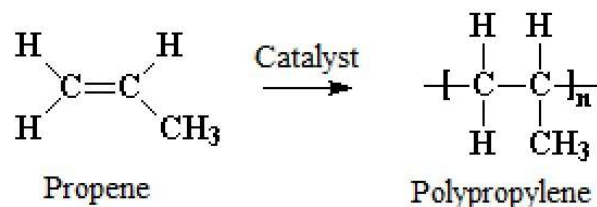
Uses:

1. These are useful in the preparation of **insulator parts, bottle caps, flexible bottles, pipes** etc.
2. LDPE is used in making film and sheeting.
3. Pipes made of LDPE are used for both agricultural, irrigation and domestic water line connections.
4. HDPE is used in manufacture of toys and other household articles.

2. Polypropylene (PP):

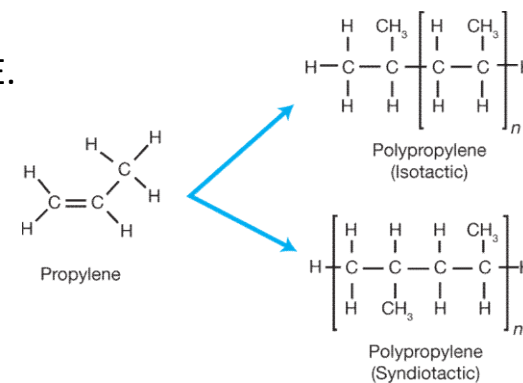
Synthesis:

It is obtained by polymerizing propylene in presence of Ziegler-Natta Catalyst ($\text{AlR}_3\text{-AlCl}_3$)



Properties:

- PP is stereoregular (isotactic) and highly crystalline polymer (mp= 160-170°C).
- It possess better hardness, strength and stiffness than PE.
- It is more resistant than PE.



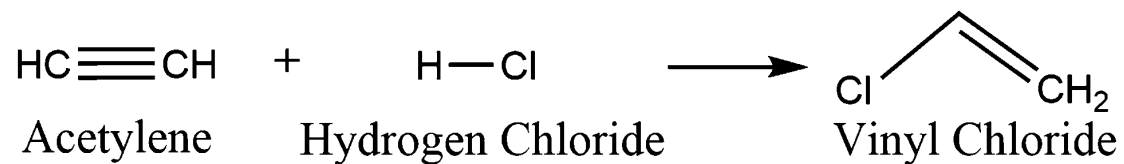
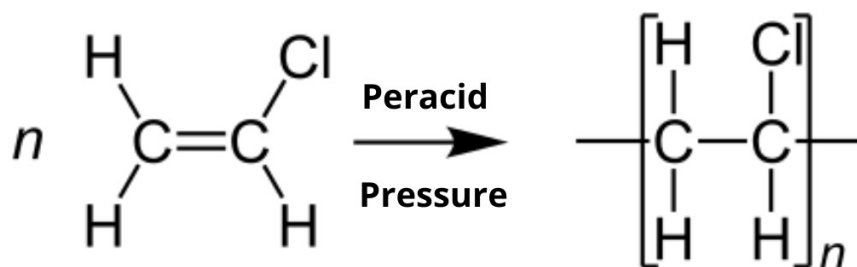
Uses:

- In producing moulded parts and fibres.
- Its fibres are used in making ropes (extremely strong by weight), carpets (indoor as well as outdoor purposes), blankets, hand bags, etc.
- Used for making water pipes, washing machine parts, serializable hospital equipment, etc.

3. Poly Vinyl Chloride (PVC):

Synthesis:

It is obtained by heating a water-emulsion of Vinyl Chloride in presence of small amount of Benzyl Peroxide or Hydrogen Peroxide in an autoclave under pressure.



Properties:

- PVC is colorless, odourless, noninflammable, and chemically inert powder, resistant to light, atmospheric oxygen, etc.
- Pure resin possesses a high softening point (145°C) and a greater stiffness and rigidity compared to PE, but is brittle.

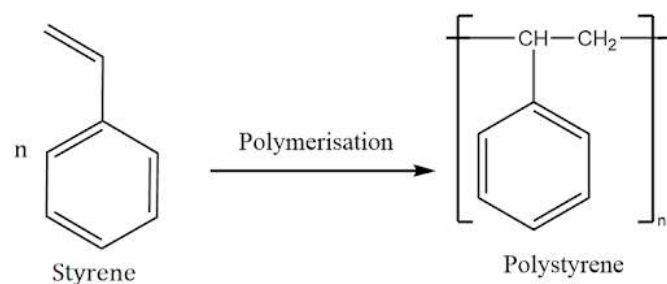
Uses:

- It is used for making sheets, which are employed for light fitting, safety helmets, motorcycle mudguards, pipes, bottle, etc.

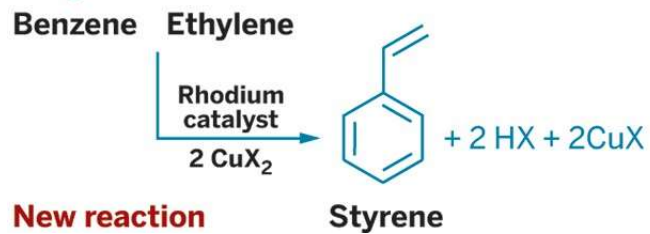
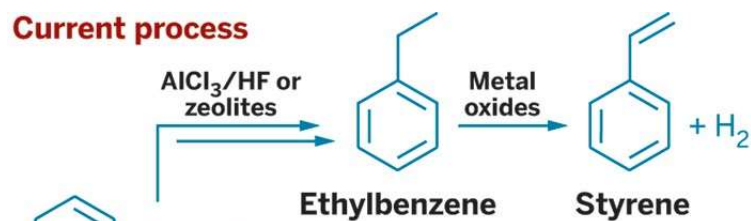
4. POLYSTYRENE (PS)

Synthesis:

It is prepared by polymerization of **styrene** (dissolved in ethyl benzene) in the presence of **benzoyl peroxide** catalyst



Current process



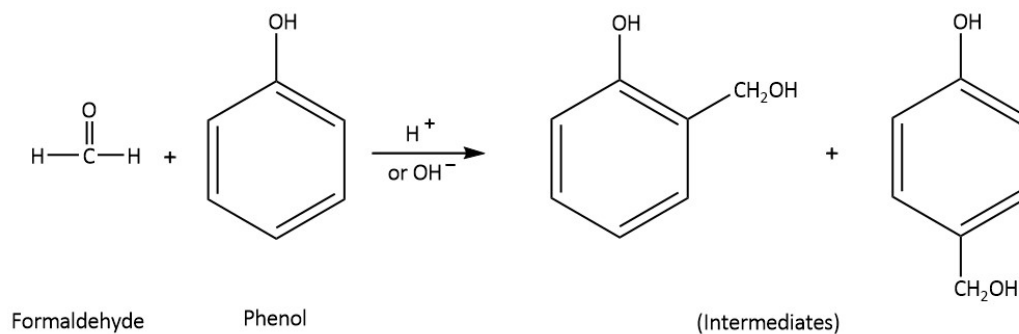
Properties:

1. PS is a **transparent, light, excellent moisture resistant**
2. It can be **nitrated** by fuming nitric acid and **sulfonated** by conc. H_2SO_4
3. At above 100°C , it yields **water-soluble emulsion**.
4. It is highly **electric insulating**, highly **resistant to acid** and good chemical resistant.
5. It has relatively **low softening range** ($90\text{-}100^\circ\text{C}$) & is brittle.

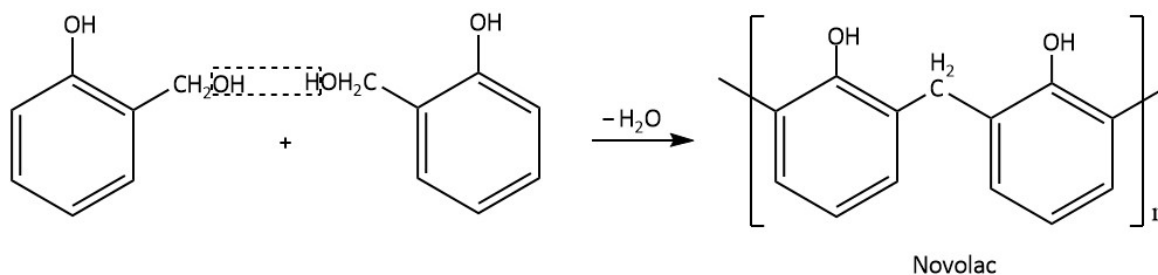
Uses:

In the molding of articles like toys, combs, radio & television parts, refrigerator parts, battery cases, high-frequency electric insulators, lenses, indoor lighting panels, etc.

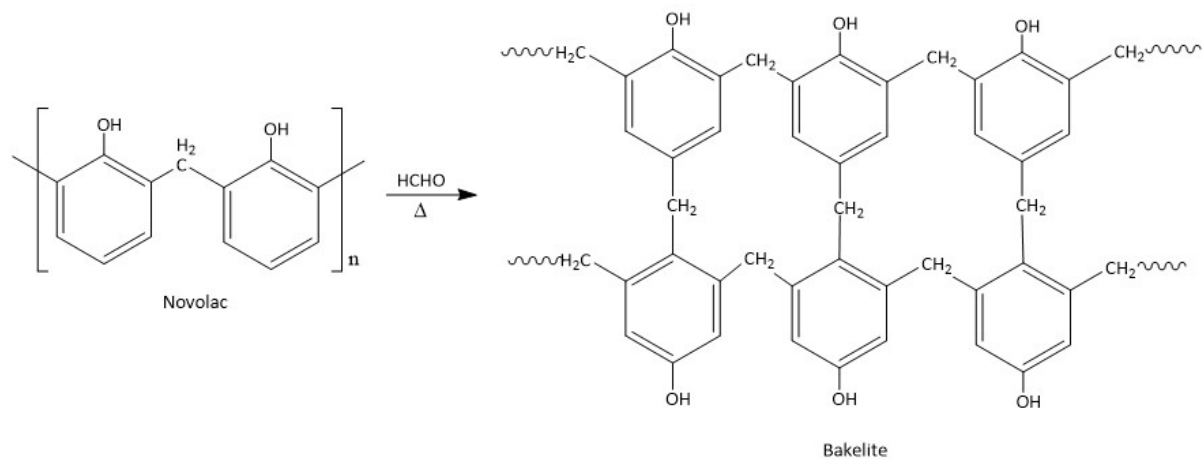
5. PHENOL FORMALDEHYDE (PF)



When two **monoethylol phenols** react with each other, Novolac resin is produced by the elimination of a water molecule. Novolac resin is also referred to as phenol-formaldehyde resin.



The **Novolac** resin is further heated in the availability of formaldehyde which leads to the formation of a cross linked polymer which is referred to as **Bakelite**.



Properties of Bakelite

- It can be molded very quickly, decreasing production time.
- Bakelite Moldings are smooth, retain their shape and are resistant to heat, scratches, etc.
- They are also resistant to several destructive solvents.
- Owing to its low electrical conductivity, bakelite is resistant to electricity.

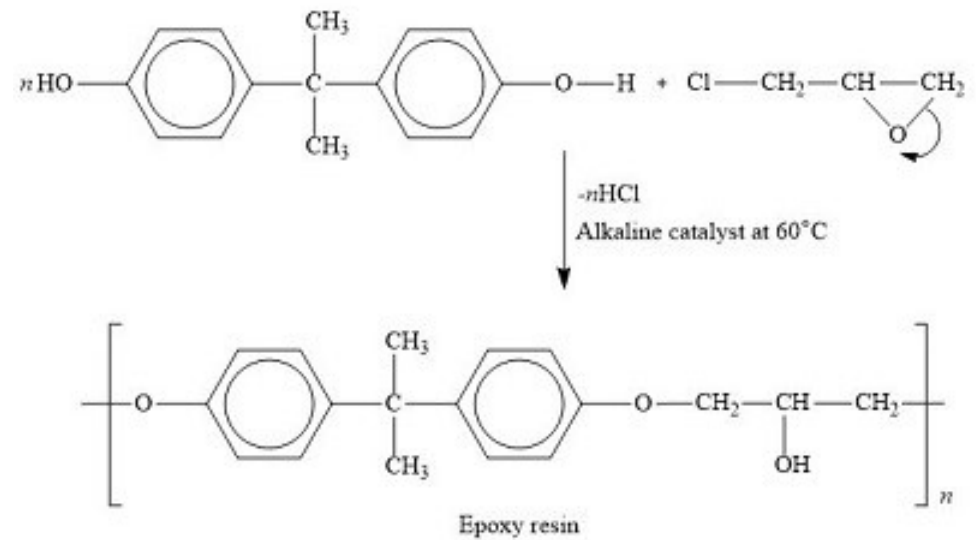
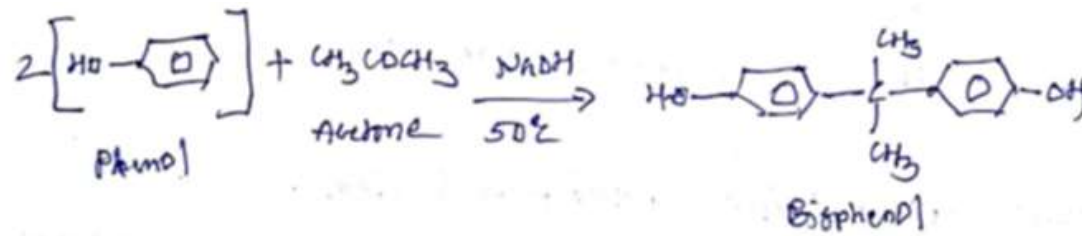
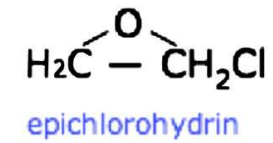
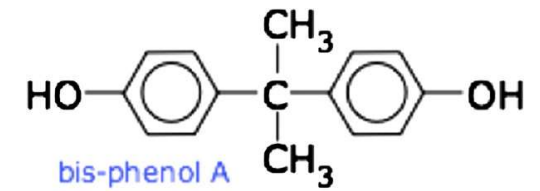
Uses:

- automotive and woodworking industries.
- **Automotive components** such: as brake pads, fuel pumps, and transmission parts are all made with phenolic-based resins.
- In the **woodworking space**, phenol formaldehyde is sometimes used in the manufacturing of medium-density **fiberboard and particleboard**. Those materials find application in **furniture making, cabinetry, and flooring products**.
- The function of the resin is to create a **strong and durable bond** between the **wood fibers**. The resulting bond is impervious to moisture and insects. It is a valuable resin that is essential to many industrial processes.

6. EPOXY RESIN

Synthesis:

It is a combination of Bisphenol and Epichlorohydrin



Properties:

- Resistance to chemicals, particularly alkaline environments.
- Heat resistance.
- Adhesion to a variety of substrates.
- High tensile, compression, and bend strengths.
- Low shrinkage during curing.
- High electrical insulation and retention properties.
- Corrosion resistance.

Uses:

- i. Paints and coatings
- ii. Adhesives
- iii. Industrial tooling and composites
- iv. Electrical systems and electronics
- v. Petroleum & petrochemical
- vi. Consumer and marine applications
- vii. Aerospace applications

3) epoxy resins are applied over cotton, rayon & bleached fabrics to impart crease-resistance & shrinkage control.

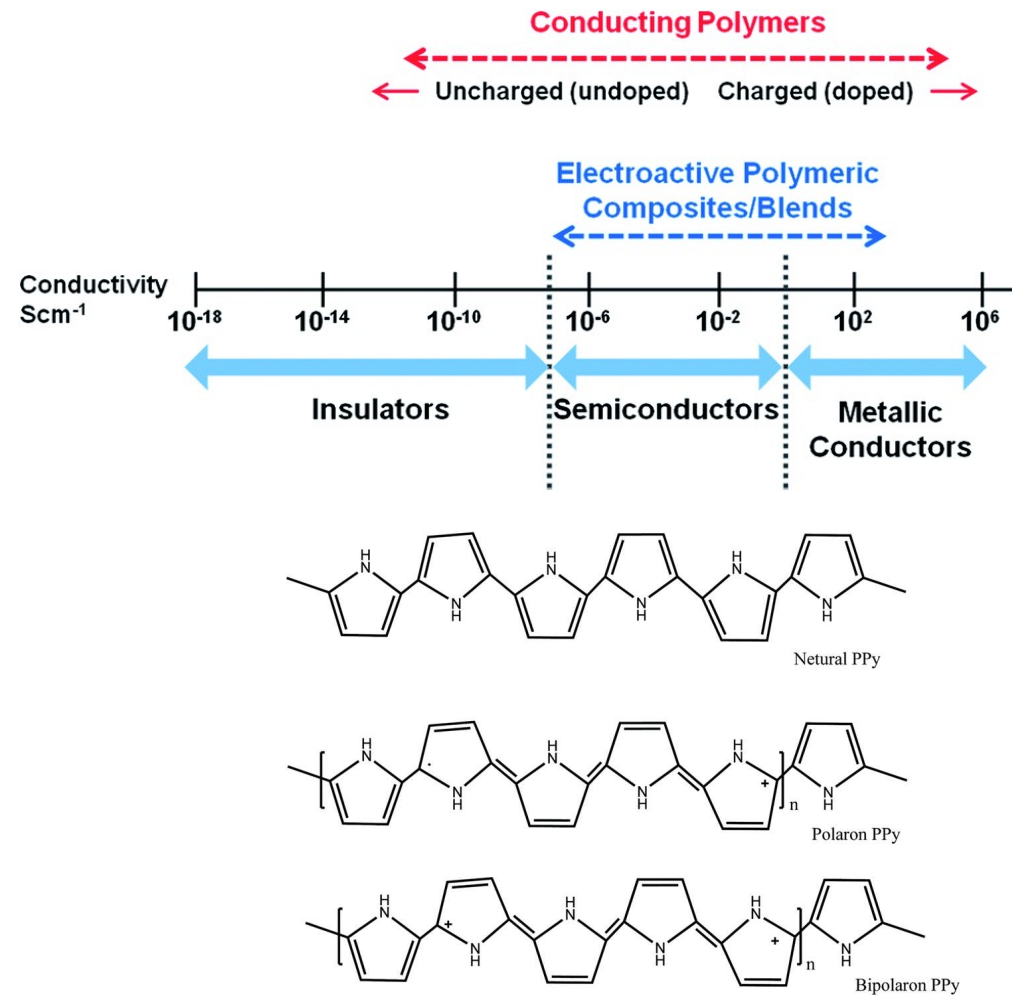
4) As laminating materials, used in electrical equipment.

Conductive Polymers:

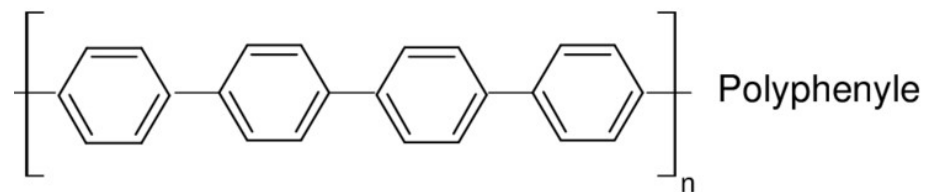
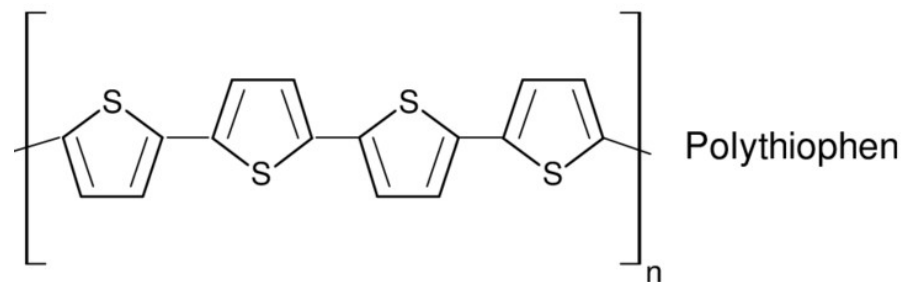
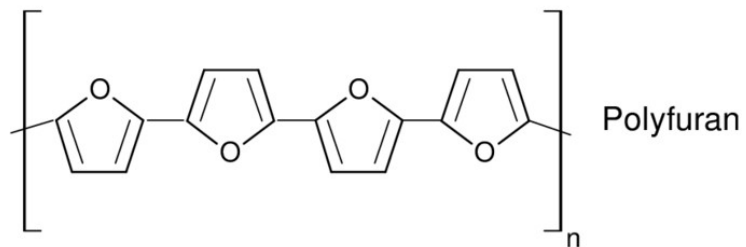
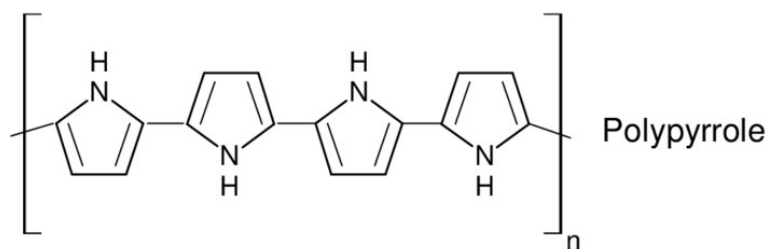
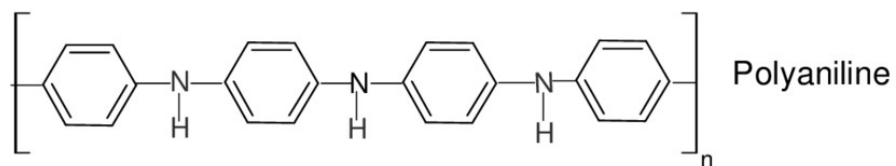
- Conductive polymers or, more precisely, intrinsically conducting polymers (ICPs) are **organic polymers** that conduct electricity.
- They have **alternating single and double bonds** along the polymer backbone (conjugated bonds).
- It is composed of **aromatic rings** such as Phenylene, naphthalene, anthracene, pyrrole, and thiophene, which are connected through carbon-carbon single bonds.
- The biggest advantage of conductive polymers is their processability, mainly by dispersion.
- The conduction of polymer may be due to unsaturation or due to the presence of externally added ingredients in them.

Conductivity

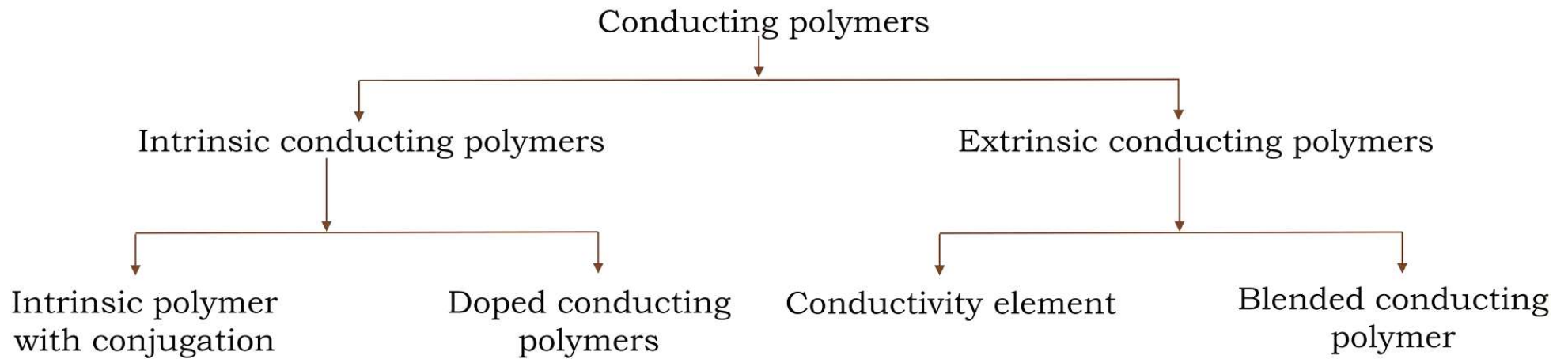
- Polymers become conducting upon **doping**
- Polymer becomes **electronically charged**
- Polymer chains generate **charge carriers**
- The concentration of **dopant** causes certain electrons to become **unpaired**
- Formation of **polarons** and **bipolarons**



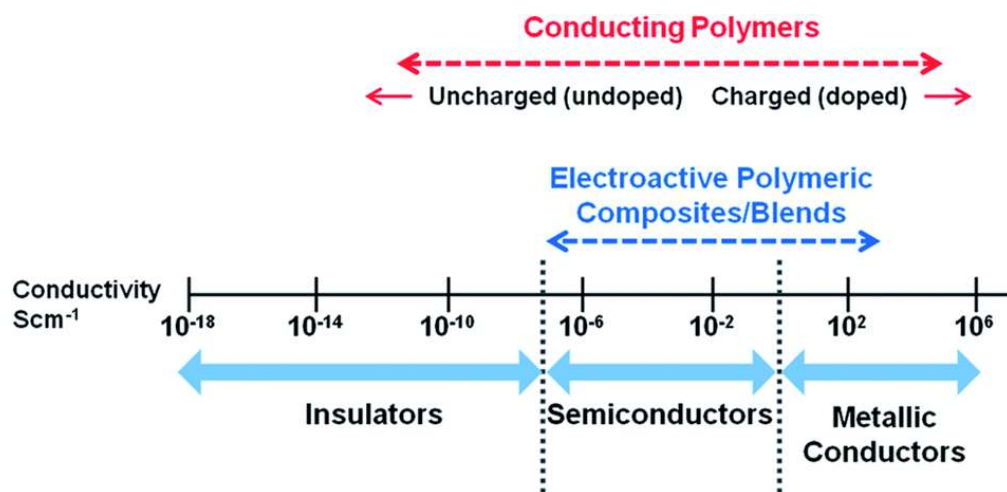
CP Examples:



Classification of conducting polymers



Intrinsic semiconductors	Extrinsic semiconductors
These are pure semiconducting tetravalent crystals.	These are semiconducting tetravalent crystals doped with impurity atoms group III (or) V
Their electrical conductivity is low.	Their electrical conductivity is high.
There is no permitted energy state between valence and conduction band.	There is no permitted energy state of the impurity atom between valence and conduction band.
Their electrical conductivity depends on temperature.	Their electrical conductivity depends on temperature as well as dopant concentration.



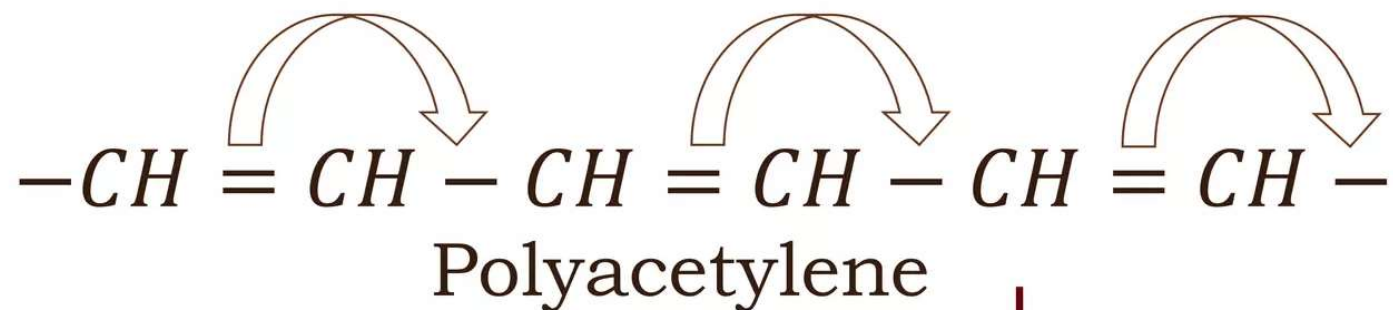
Intrinsically conducting polymers are substances which have a π -bond backbone.

- There are certain electrons which are extra in this type of polymers.
- These extra electrons flow from one point to another in the polymer, as a result they have the ability to conduct electricity.
- **Conduction of electricity in this type of polymers is due to conjugation in the backbone of polymer.**
- The conjugation can be due to either electrons or due to doped ingredients.

Conduction due to conjugated π electrons:

- The presence of **double bonds and lone pair of electrons** conduction of electricity takes place.
- Due to the **overlapping of conjugated π electrons**, valence and conduction bands throughout the backbone of the polymer are developed.
- **Electrical conduction** can occur only after the attainment of the required **energy of activation**, either **thermally** or **photochemically**, because there is some **gap** between the valence and conduction bands.
- **Polyacetylene, polyaniline**, etc., are these types of conducting polymers.

Example:



Delocalisation of
 π electrons



Doped Conducting Polymers

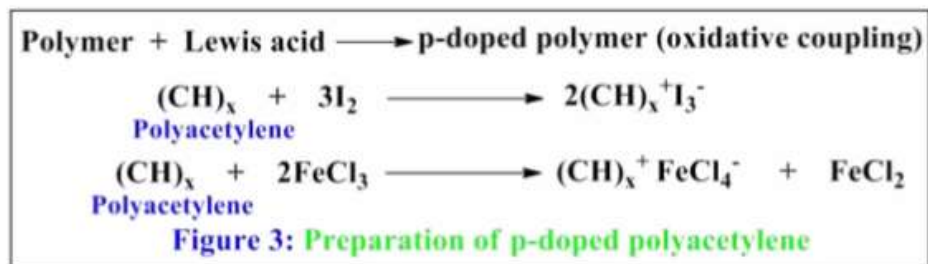
- The conduction power of semiconductor can be enhanced by adding some **foreign material** or **desired impurities**.
- **These impurities are called doping agent or dopant.**
- **Appropriate doping agent increase the conductivity of semiconductors up to 10^4 times.**
- The increase in conduction is due to the **participation of impurity elements** in between the valence band and conduction band.
- Making a **bridge** through which electrons can jump easily from the valence band to the conduction band.

The foreign materials develop **positive or negative charge** through oxidation or reduction of the semiconductor.

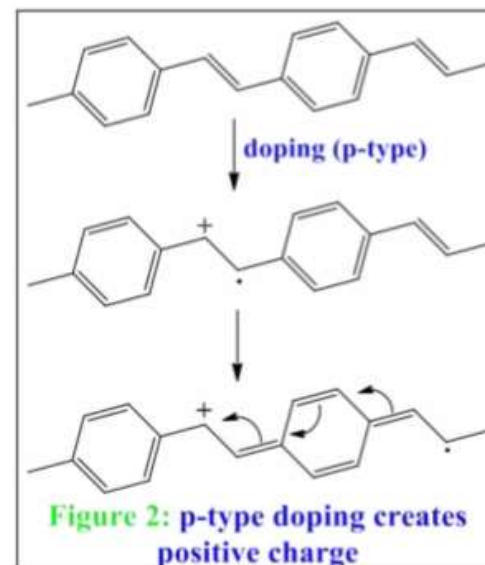
Doping are mainly two types.

1. p-type doping through oxidation of materials:

- Some electrons from the conjugated π bonds are removed through oxidation creating a **positive hole** called **polaron** inside the polymer.
- The **positive hole or polaron** can move throughout the polymeric chain and make it conducting polymer.

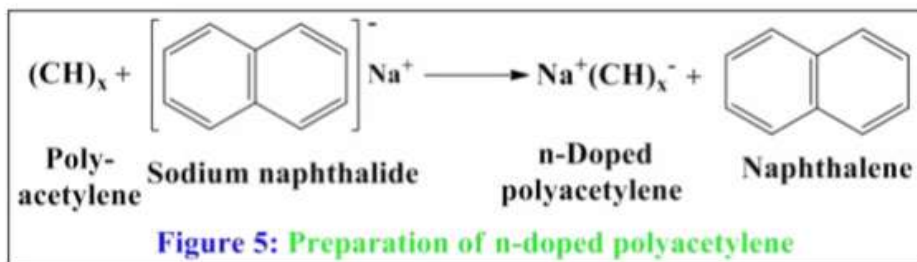


The polymers which have conjugation in the backbone when treated with electron-deficient species (Lewis acid) like FeCl_3 or I_2 vapour or I_2/CCl_4 , oxidation takes place and a positive charge is created in the molecule.

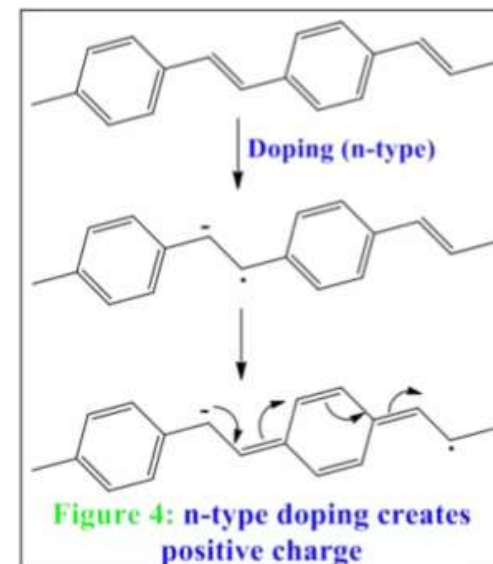


2. n-type doping through reduction of materials:

- Some electrons are introduced to the conjugated π bonds through reduction creating a **negative hole or charge** inside the polymer.
- The negative hole or charge can move throughout the polymeric chain and make it conducting polymer.
- Lewis bases, $\text{Na}^+\text{C}_{10}\text{H}_8^-$, $\text{K}^+\text{C}_{10}\text{H}_8^-$, etc., are generally used as doping agents.



When Lewis bases (electron rich species) are treated with polymer having conjugation, due to reduction of the polymers, negative charge develops.



Extrinsically Conducting Polymers (ECPs)



- Those **conducting polymers** which owe their conductivity due to the presence of externally added ingredients in them are called **extrinsically conducting polymers**. Extrinsically conducting polymers (ECP's) are of two types. These are: (1) conducting elements filled polymers (CEFP) i.e., the polymers filled with conducting element, and (2) blended conducting polymers (BCP).
1. **Conducting Elements Filled Polymers (CEFP):** In this type, a conducting element is added to the polymer. **Therefore, the polymer acts as a binder to hold the conducting elements together in solid entity.** Thus, conductivity of these polymers is due to the addition of external ingredients. Upon addition of conducting element, the polymer will have a property of that conducting element and it will start conducting electricity.
 2. **Blended conducting polymers:** These types of polymers are obtained **by blending a conventional polymer with a conducting polymer either physically or chemically.** This blend of polymers conduct electricity. Such polymers can be easily processed and possess better physical, chemical and mechanical properties.

Applications of conducting polymers:

1. In rechargeable batteries.
2. In making analytical sensors for pH, O₂ , SO₂ , NH₃ , glucose, etc.
3. In bio-medical applications.
4. In controlled release of drugs.
5. In optical filters.
6. In photo voltaic devices.
7. In telecommunication systems.
8. In micro-electronic devices.

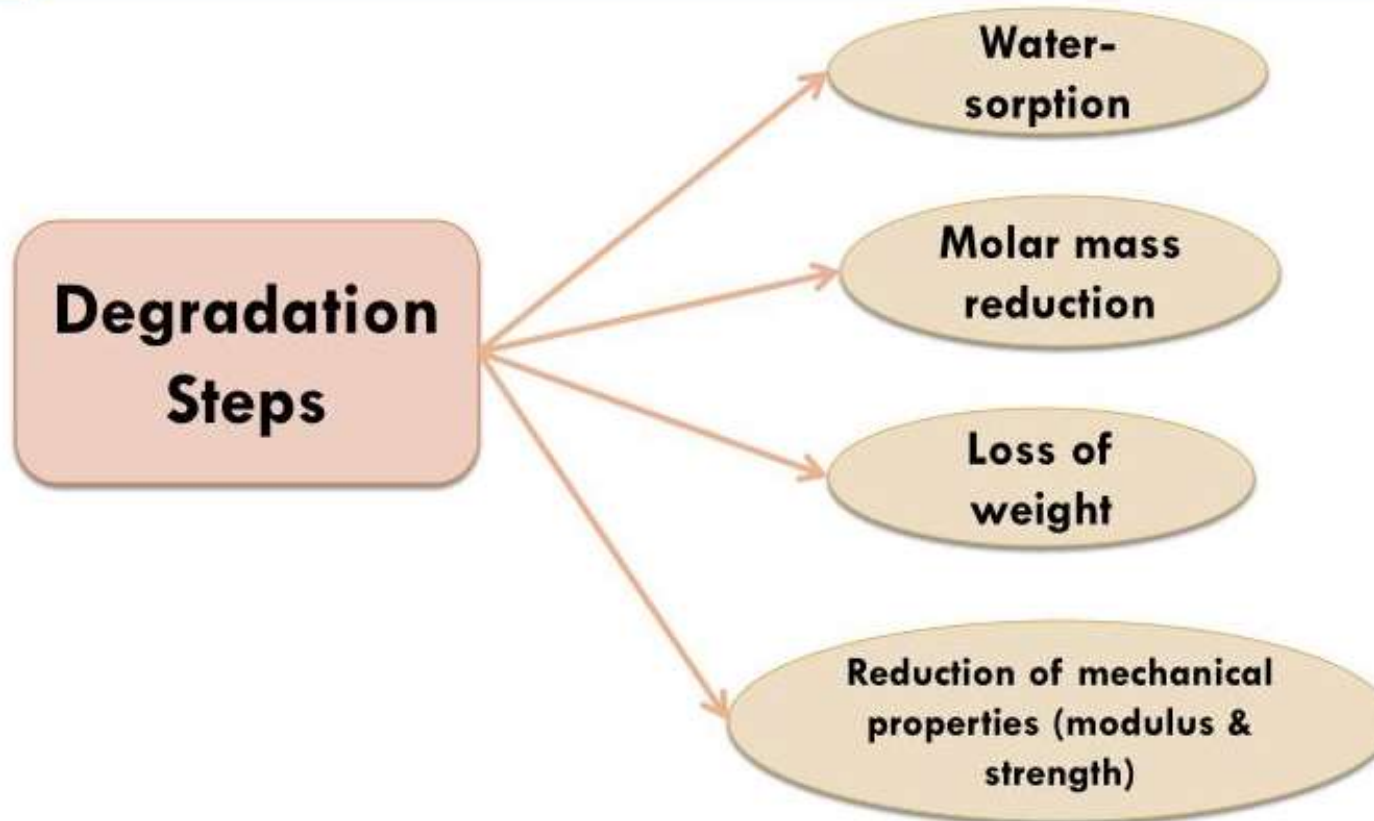
Biodegradable Polymers

- Polymers which get degraded into simple molecules like CO₂, water, methane, inorganic compounds, biomass, etc. under aerobic or anaerobic environment by the action of microorganisms or enzymes.
- Polymers comprised of monomers linked to one another **through functional groups** and have **unstable links** in the backbone

Polymer Degradation

- ▶ Polymer degradation is a change in the properties – tensile strength, colour, shape, etc of a polymer or polymer based product under the influence of one or more environmental factors such as heat, light or chemicals.

Degradation Steps



Factors affecting Biodegradation of Polymers

➤ Morphological factors

- Shape & size
- Variation of diffusion coefficient and mechanical stresses

➤ Chemical factors

- Chemical structure & composition
- Presence of ionic group and configuration structure
- Molecular weight and presence of low molecular weight compounds

➤ Physical factors

- Processing condition
- Sterilization process

Based on biodegradability polymers are classified as:

1. Biodegradable polymers

e.g.: poly glycolic acid, Poly lactic acid etc.

2. Non biodegradable polymers

e.g.: poly vinyl chloride, polyethylene etc.

Classification of biodegradable polymers based on the source

1) Synthetic biodegradable polymers:

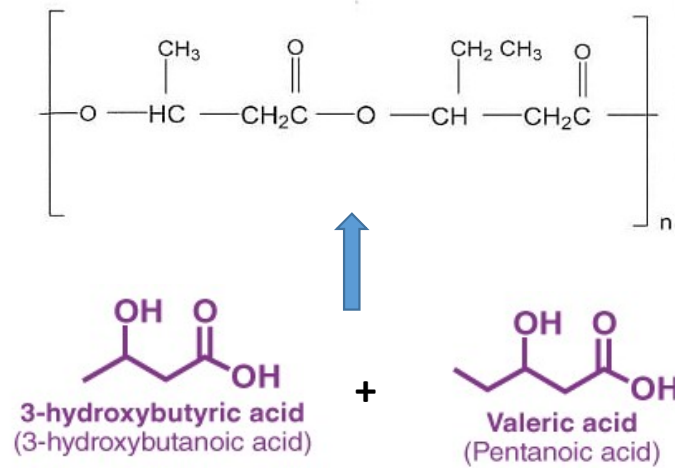
e.g.: Aliphatic poly(esters), Polyanhydrides, Polyphosphazenes, polyaminoacids, etc.

2) Natural biodegradable polymers:

e.g: Albumin, Collagen, Dextran, Gelatin, starch etc.

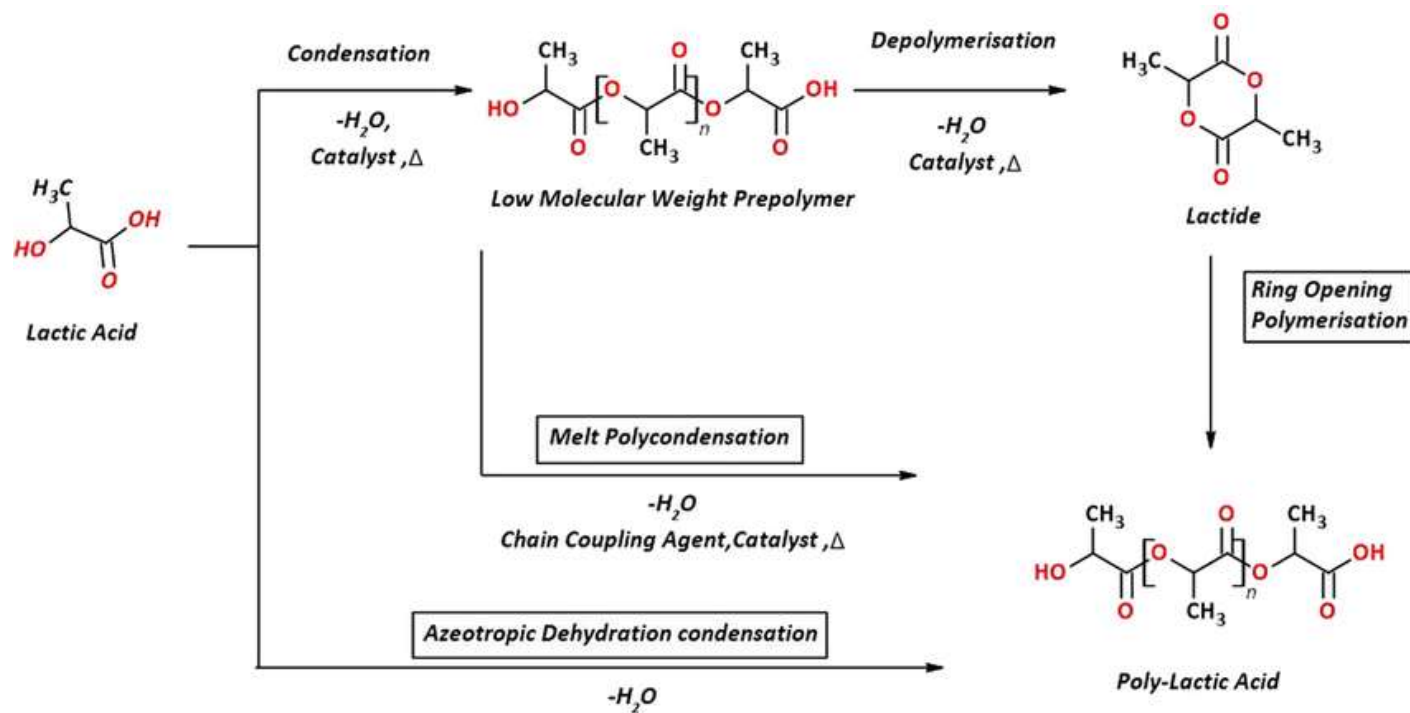
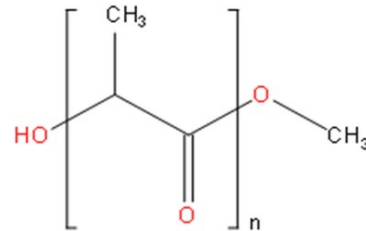
Examples:

1. Poly-3-hydroxyl butyrate-co-3-hydroxyl valerate (PHBV)

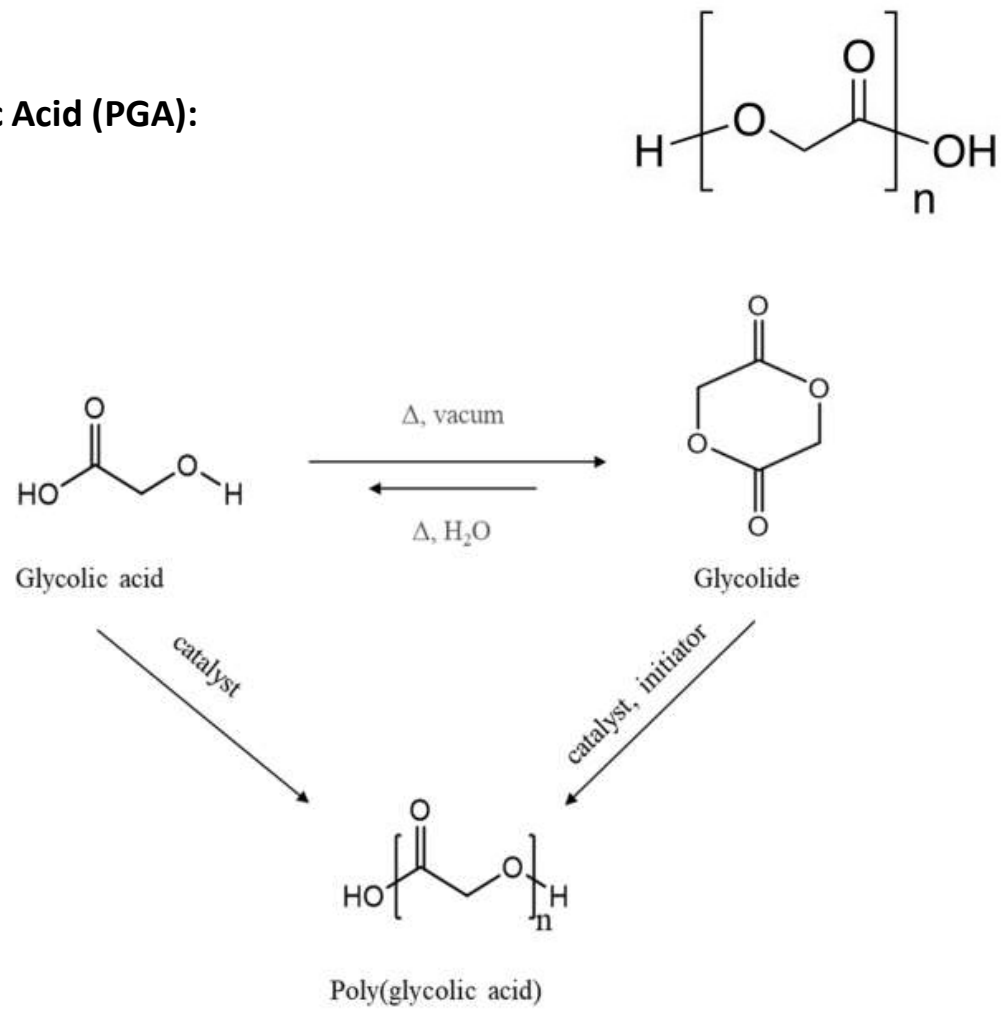


2. Polylactic acid (PLA):

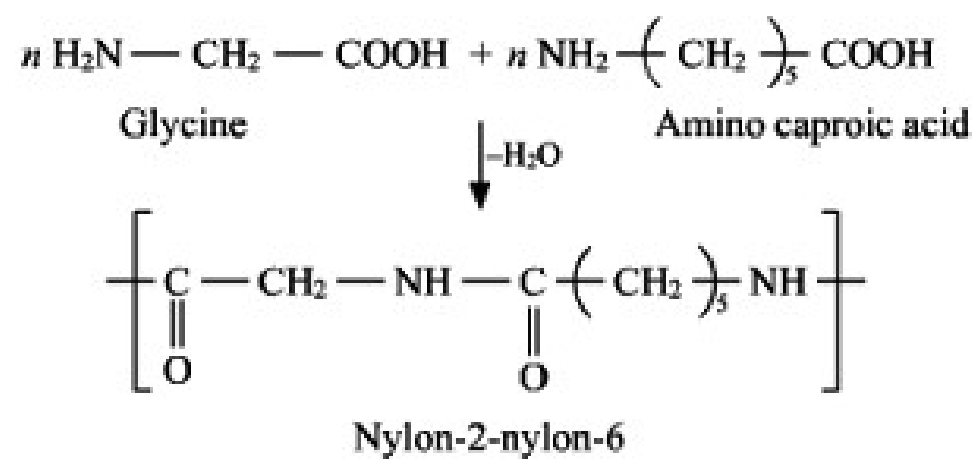
Synthesis:



3. Polyglycolic Acid (PGA):



4. Nylon-2 Nylon-6



Properties:

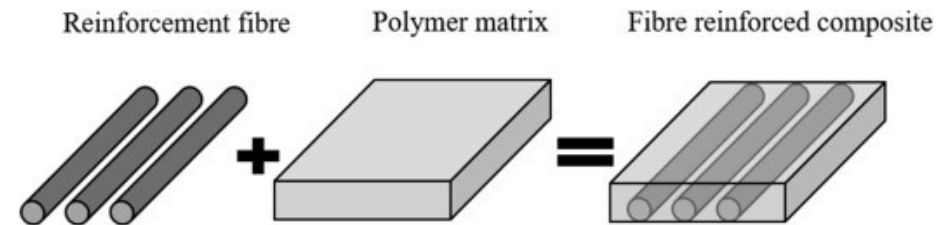
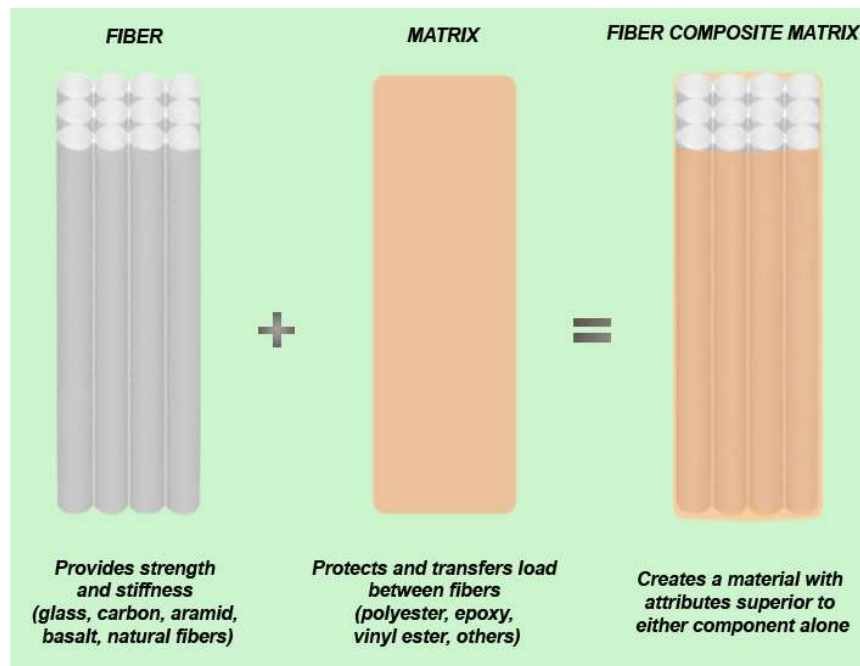
- Degradation usually starts at the end groups because BDP have exceptionally strong carbon backbones that are difficult to crack.
- BDP can maintain strong mechanical integrity until they are degraded.
- Non-toxic
- The degradation rate can be controlled
- Hydrophilic polymers

Uses:

- Used for stitches after surgery
- Utilized in agriculture goods like seed coating and films
- In gene therapy
- These are employed in medicinal agents such as anticancer, antipsychotic, and anti-inflammatory drugs in the biodegradable system.
- Fast-food packaging & personal hygiene items.

Composites

- A composite material (composites) is a material that consists of two or more discontinuous components (particles/fibers) that are placed in a continuous medium (matrix).



Characteristics:

- High specific strength
- High specific stiffness
- Low Density
- Unique combination of properties
- Corrosion resistance

- In a fiber composites the matrix binds together the fibres, transfer loads between the fibers and protects them from environment and external damage.

Examples:

1. **Wood:** Cellulose fibers bound by lignin matrix
2. **Granite:** Quartz, feldspar and mica
3. **Concrete:** Cement, water, sand, gravel

Constituents of Composites:

1. **Matrix Phase :** Continuous body constituent which encloses the composite and gives its bulk form e,g, polymer
2. **Dispersed phase:** Structural constituent which determines internal structure of the composite. Fibres, particulate, flakes, etc.

Composites

Particulate reinforced

Large
Particle

Dispersion
Strengthened

Fiber
reinforced

Continuous Discontinuous

Aligned

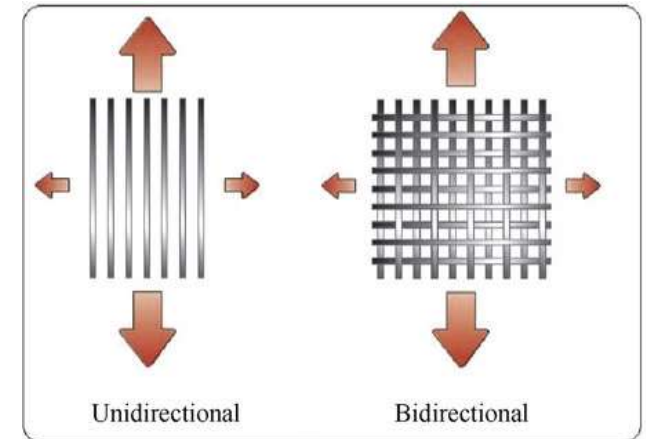
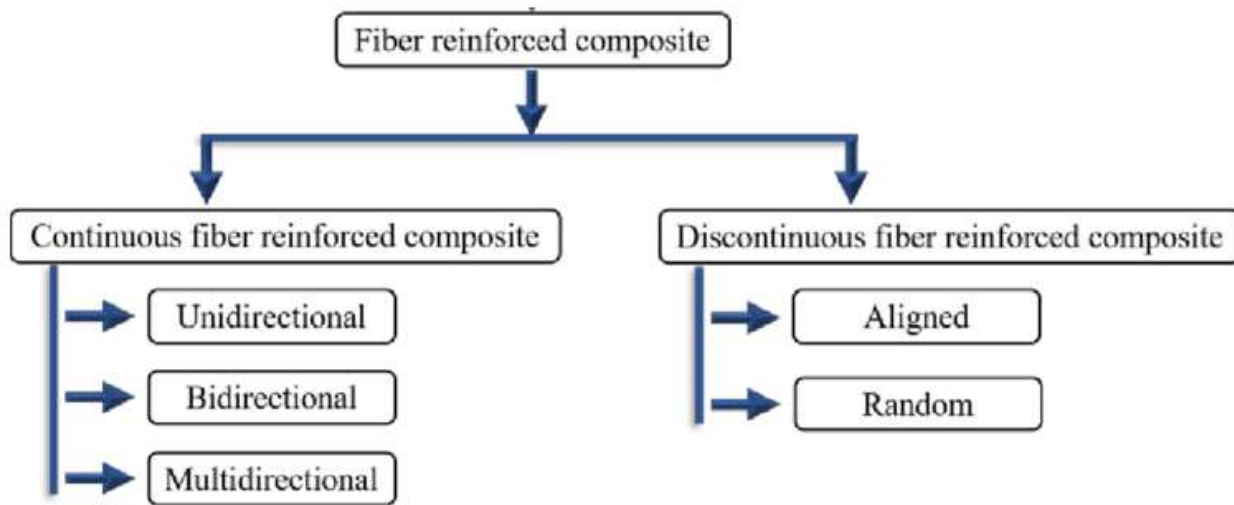
Random

Structural

Laminates

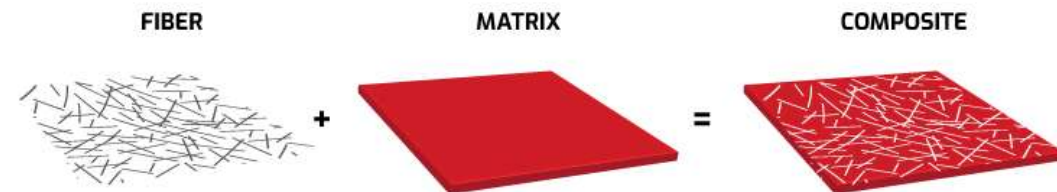
Sandwich
Panels





A. Fiber-reinforced composites (FRC):

- Dispersed phase: fiber e.g. carbon, glass, etc.
- Continuous phase: polymer.



Components of composite materials

Reinforcement: fibers

Glass
Carbon
Organic
Boron
Ceramic
Metallic



Matrix materials

Polymers
Metals
Ceramics



Interface

Bonding
surface

- FRC possesses high specific strength, stiffness, corrosion resistance and lower density

The mechanical characteristics of FRC depend on:

- Properties of fiber
- Interfacial bond between fiber and matrix
- Fiber length like longer gives continuous, and shorter length gives discontinuous or random.
- The reinforcement efficiency of continuous fibers is higher than short fibers
- Fiber orientation & concentration i.e. if it is orderly oriented and continuous it is highly anisotropic or discontinuous or random orientation

General Properties of Composites

- High Strength to weight Ratio
- Lightweight
- Fire Resistance
- Electrical Properties
- Chemical and weathering resistance
- Color
- Translucency
- Design
- Low Thermal Conductivity Manufacturing Economy

Examples:

1. **Glass fiber reinforced polymer composites:** glass fiber + plastic matrix
2. **Carbon fiber reinforced polymer composites:** carbon fiber like graphite, graphenes, CNTs + polymer matrix
3. **Alumina Oxide/carbon fiber reinforced metal composites:** Fiber of alumina (Al_2O_3)/Carbon in metal or metal alloy matrix

B. Particle (particulate) reinforced composites (PRC)

Solid particles of metal oxides or carbides of varying sizes are dispersed in metal, metal alloy, ceramic, or polymer matrix.

PRC is classified into two types:

1. Large particulate composites
2. Dispersed strengthened composites

1. Large particulate composites:

It is used with all three major types of materials: metal, polymer, and ceramics.

Examples:

1. **Concrete:** Cement matrix and particulates of sand and gravel
2. **Automobile tire:** carbon black particles dispersed in rubber matrix

2. Dispersed strengthened composites

- Very small particles in the range of 10-100nm size are used.
- It improves strength and hardness.
- **Example:** Thoria-dispersed nickel: used in various high temp. operations like combustion engines

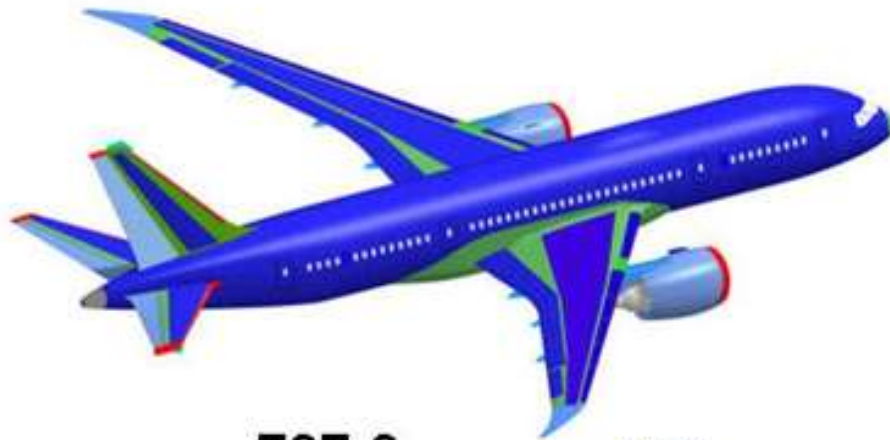
C. Structural Composites:

- prepared by compressing the stacking of layers of fiber-reinforced composites.
- **Types:** Laminated and sandwich

Applications of Composite Materials

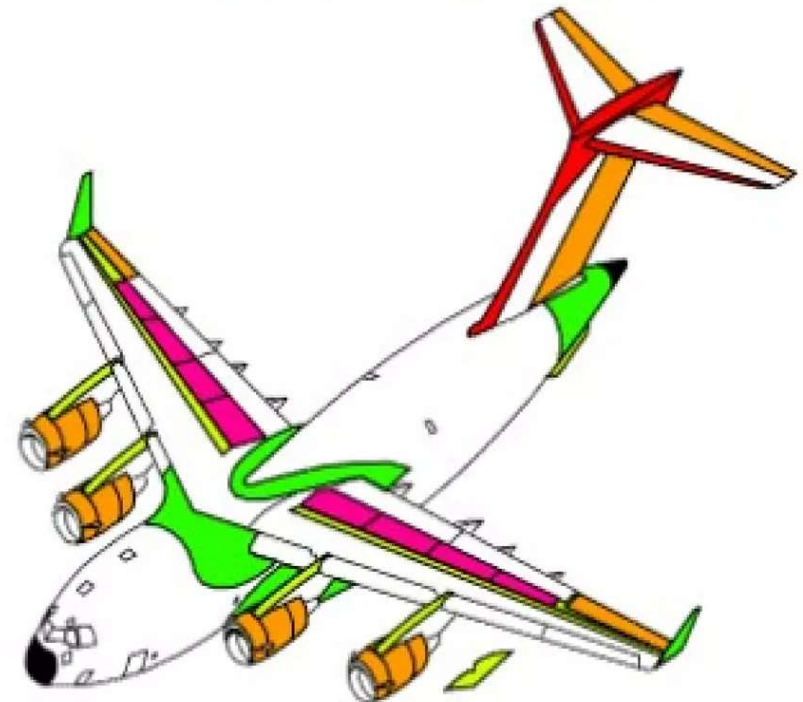
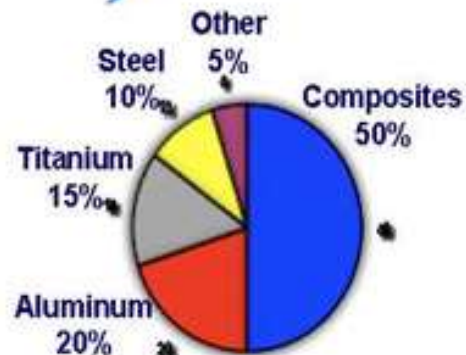
1. Aircraft Industries

C-17 Aircraft



787-8

- Carbon laminate
- Carbon sandwich
- Other composites
- Aluminum
- Titanium



- Carbon/epoxy
- Aramid/DuPont Nomex
- Carbon/aramid/epoxy
- Aramid/foam core
- Glass-fiber reinforced plastic
- Carbon/DuPont Nomex

Applications of Composite Materials

1. **Aeronautical applications:** Structural components like wings, body & stabilizer & fuel of aircraft, rocket army missiles in military, etc.
2. **Automobile industries:** Automobile parts like components of the engine, spray nozzle, mudguards, tires, etc.
3. **Communication industry:** preparation of antennae and electronic circuit boards.
4. **Safety equipment:** helmets.
5. **Marine applications:** Shaft, hulls, and other parts of ships.
6. **Sports equipment:** tennis rockets and golf sticks.



Disadvantages & Limitations of Composite Materials

- Properties of many important composites are **anisotropic** - the properties differ depending on the direction in which they are measured – this may be an advantage or a disadvantage
- Many of the polymer-based composites are subject to **attack by chemicals or solvents**, just as the polymers themselves are susceptible to attack
- Composite materials are generally **expensive**
- Manufacturing methods for shaping composite materials are often **slow and costly**